

RESEARCH

Open Access



A record-setting mitogenome in the holoparasitic plant *Balanophora yakushimensis* accompanied by exceptional loss of organellar DNA repair and recombination genes

Runxian Yu^{1,9†}, Xueke Zhi^{1†}, Luis Federico Ceriotti^{2†}, Elizabeth Skippington^{3,10†}, Danny W. Rice³, Huei-Jiun Su⁴, Todd J. Barkman⁵, Chenyu Sun¹, Ying Liu¹, Dongming Fang⁶, Xiaoli Chen⁶, Claude W. dePamphilis⁷, Jeffrey P. Mower⁸, M. Virginia Sanchez-Puerta^{2*} , Jeffrey D. Palmer^{3*} and Renchao Zhou^{1*}

Abstract

Background Despite only limited sampling, the holoparasitic plant family Balanophoraceae harbors extreme mitogenome diversity and also has exceptionally divergent plastomes. We therefore sequenced the mitochondrial, plastid, and nuclear genomes of *Balanophora yakushimensis* and its transcriptome.

Results At 1.1 Mb, the *B. yakushimensis* mitogenome is one of the largest known mitogenomes. Driving this expansion and generating the most repeat-rich mitogenome in land plants are many large (up to 200 kb) duplications and a massive proliferation of short, AT-rich repeated sequences. The repeat proliferation, in conjunction with a highly elevated and unusually AT-biased mutation rate, has produced what is by far the most AT-rich land-plant mitogenome. These invasive repeats also created giant introns, unprecedented in size for organelles, and greatly expanded all rDNA exons. We discovered a record-low, for *all* genomes, transition/transversion ratio (0.12) in *B. yakushimensis* mtDNA and documented a 26-fold range in this ratio across angiosperm mitogenomes. The *B. yakushimensis* nuclear genome has lost exceptionally many genes that function in organellar DNA recombination, repair, and replication (RRR). We discuss ways in which these losses—and other genetic alterations as well as non-genetic ones—may or may not be related to the unusual features of both its mitochondrial and plastid genomes.

Conclusions The mitogenome of *B. yakushimensis* possesses many exceptional, indeed record-setting properties. The unprecedented loss of nuclear genes for organellar DNA RRR may explain some of these unusual features. These findings expand the boundaries of mitogenome deviancy and raise outstanding questions about the forces driving such extravagantly diversifying evolution.

Keywords Mitochondrial genomes, Mutation spectrum, Repeated sequences, AT-biased base composition, Photolyases, Parasitic plants

[†]Runxian Yu, Xueke Zhi, Luis Federico Ceriotti and Elizabeth Skippington contributed equally to this work.

*Correspondence:
M. Virginia Sanchez-Puerta
mvsanchezpuerta@fca.uncu.edu.ar
Jeffrey D. Palmer
jpalmer@iu.edu
Renchao Zhou
zhrench@mail.sysu.edu.cn
Full list of author information is available at the end of the article



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Background

Mitogenomes of land plants are famous for their evolutionary volatility, exhibiting extreme variation in numerous properties including genome size, mutation rate, structure, gene content, and levels of horizontal gene transfer (HGT) [1–5]. Genome copy-number contributes to some of the disparate properties as does the loss of nuclear genes for organellar DNA replication, repair, and recombination (DNA RRR) and mutations to critical sites of the encoded proteins (e.g. [5–10],).

Certain holoparasitic angiosperms—nonphotosynthetic plants that parasitize other plants—possess among the most divergent and therefore exciting mitogenomes. Indeed, two of the ca. 15 evolutionary lineages of holoparasitic angiosperms stand out not only for their remarkable mitogenomes, but also for their aberrant plastid and nuclear genomes, as well as their bizarre morphologies. One lineage is the Rafflesiaceae, a tropical family of obligate endoparasites that lack leaves, stems, and roots, yet produce the world's largest flowers. While almost all angiosperm holoparasites have lost most plastid genes [4], Rafflesiaceae has lost its entire plastome [11, 12], making it one of only three land-plant groups thought to have gone this far [13, 14]. Rafflesiaceae mitogenomes have two unusual features, high levels of HGT, and a highly multi-circular structure [11, 15–17]. The nuclear genome from *Sapria himalayana* is exceptionally derived, with massive loss of native genes, many acquisitions of foreign genes, and highly unusual introns [12, 16].

The second standout lineage of holoparasites is the Balanophoraceae, a largely tropical family of highly reduced and morphologically bizarre root parasites, with some of its 53 species mistaken for fungi because of their fleshy, mushroom-like inflorescences [4, 18]. In striking contrast to Rafflesiaceae, *Balanophora* possesses some of the smallest flowers known, with as many as 1 million produced per female inflorescence [18, 19].

Despite only limited sampling, the Balanophoraceae harbors some of the most amazing mitogenomes known, in plants or otherwise. The *Rhopalocnemis phalloides* mitogenome is extremely small with tiny introns, has converged on a minicircular organization found otherwise only in several animals and protists in which all 21 minicircles contain an identical putative DNA-replication origin, and displays an unprecedented level of heteroplasmy, leading to a plethora of expressed sequence variants [8]. The relatively large mitogenomes of *Lophophytum mirabile* and *Ombrophytum subterraneum* are remarkably rich in foreign mitochondrial DNA (mtDNA) acquired by HGT, with most of this HGT occurring independently in the two species. Over 90% of the *L. mirabile* mitochondrial pangenome was acquired from legume

host plants, making it proportionately the most HGT-rich genome in any eukaryote [20, 21]. Moreover, this mitogenome stands alone among all genomes (including bacterial) in having functionally replaced most native genes by foreign homologs [22]. Finally, mtDNAs of all five examined Balanophoraceae species share one exceptional feature: They consist of many (18–81) autonomous circular chromosomes [8, 17, 23, 24].

The plastid genomes (plastomes) of Balanophoraceae are extraordinary too. All 23 sequenced Balanophoraceae plastomes from 10 genera are highly reduced in gene content and size, unusually compact, extraordinarily AT-rich, and highly divergent in sequence [25–27]. Two plastid lineages within Balanophoraceae have evolved modified genetic codes, the only code changes identified in any land-plant genome, plastid or otherwise. These code changes may be a consequence of the super-high AT content and highly reduced gene content of the affected lineages.

The two sequenced nuclear genomes from Balanophoraceae (both in *Balanophora*) have also sustained massive gene loss, with many parallel losses shared between these genomes and the genome of the holoparasite *Sapria* [12, 16, 28]. Of relevance here, *Rhopalocnemis* and *Lophophytum* appear to have lost several nuclear genes involved in organellar DNA RRR [6–8]. These apparent losses may be causally related to some of the highly aberrant features of their organellar genomes.

With so many exceptional and diverse features in Balanophoraceae mitogenomes, the family is an outstanding group in which to further investigate mitogenome evolution and potentially related losses of DNA-RRR genes. We therefore sequenced all three genomes and the transcriptome of *B. yakushimensis* and generated a draft mitogenome for *B. laxiflora*. Compared to other Balanophoraceae mtDNAs, *Balanophora* has a wholly different suite of exceptional mtDNA features, most of them record setting. Furthermore, it has lost many additional DNA-RRR genes. These findings raise exciting questions about the limits, if any, to deviant evolution in mtDNA and of the forces driving this evolution.

Results

The *Balanophora yakushimensis* mitogenome is large, highly repetitive, and complex

A combination of PacBio and Illumina sequencing resulted in a 1,141,990-bp assembly of the *B. yakushimensis* mitogenome. The assembly consists of 21 “regions,” ranging in size from 282 to 199,962 bp (sum = 645,144 bp) (Additional file 1: Table S1), that vary by eightfold in copy number and are connected in a highly complicated manner (Fig. 1A) owing to recombination between members of each of the 13 multi-copy

regions. Resolution of all pathways in Fig. 1A generates the six-chromosome model for the mitogenome shown in Additional file 2: Fig. S1. Other configurations of the *B. yakushimensis* mitogenome undoubtedly exist via recombination occurring at other sets of multi-copy regions. Owing to the many, often large repeated regions, 19 of the 38 *B. yakushimensis* genes are present in multiple copies, either entirely (13 genes) or in part (six), i.e., with part of a gene residing in a single-copy region and the rest in one or more multi-copy regions (Fig. 1B, Additional file 1: Table S2 and Additional file 2: Fig. S1).

In addition to its 13 multi-copy regions, the *B. yakushimensis* mitogenome contains upwards of 47,000 additional repeat families (Fig. 1B, Additional file 1: Table S3 and Additional file 2: Fig. S2). In sum, these families contain over half a million individual repeated sequences; however, the aggregate length of these repeats is “only” 444,405 bp, reflecting extremely high levels of overlap among repeat families. Most repeats are short; 92.2% are less than 50 bp in length, and 99.6% are less than 100 bp (Additional file 2: Fig. S2). Repeats comprise 68.9% of the 645,144 bp of the mitogenome assembly consisting of the 21 regions, each counted only once. When the 13 repeated regions are included, the percentage of the entire, 1,141,990-bp mitogenome covered by repeated sequences increases to 91.2%.

Illumina-only sequencing produced a draft assembly for *B. laxiflora* of 15 mitochondrial contigs, with a total length of 356,848 bp. Although incomplete, this assembly contains a complete set of mitochondrial genes relative to *B. yakushimensis* (Additional file 2: Fig. S3 and see below). Depth of coverage among and within contigs is variable (Additional file 2: Fig. S4), suggesting the existence of large repeated regions, as in *B. yakushimensis*.

We also assembled the 14,624 bp plastome of *B. yakushimensis* (Additional file 2: Figs. S5 and S6). In all respects, this genome is highly similar to those of other *Balanophora* species [25, 26].

Gene content and expression

The *B. yakushimensis* and *B. laxiflora* mitogenomes share the same set of 34 intact protein genes, three rRNA genes, and one tRNA gene. The protein and rRNA gene sets are similar to those of the four other sequenced *Balanophoraceae* mitogenomes, with the caveat that most

native protein genes in *Lophophytum* have been replaced by foreign homologs (Additional file 2: Fig. S7) [8, 20, 22–24, 29, 30]. Phylogenetic analysis (Additional file 2: Fig. S8) [23] failed to find evidence for a foreign origin of any mitochondrial protein gene in the three *Balanophora* species or in *Rhopalocnemis* and *Thonningia* (Additional file 3: A) [1, 4, 8, 24]. Nor did we find any evidence for intracellular gene transfer (IGT) in *B. yakushimensis* mtDNA (Additional file 3: A). Transcription of all protein and rRNA genes was verified in *B. yakushimensis* by RNAseq, with C-to-U mRNA editing occurring predominantly, as expected, at nonsynonymous sites and at a level comparable to that of other eudicots (Additional file 1: Table S4) [31].

The complement of native tRNA genes is highly variable in *Balanophoraceae*, ranging from 0 to 10 (Additional file 1: Table S5 and Additional file 2: Fig. S7). *Balanophora* and *Lophophytum* have independently reduced to one and zero native tRNA genes, respectively. These are the smallest tRNA gene sets in angiosperms, rivaled only by those in *Viscum* and certain *Silene* species [32, 33].

Enormous introns

The *B. yakushimensis* mitogenome contains 24 group II introns and one group I intron (Additional file 1: Table S6) [34–36], all of which are spliced properly based on the RNAseq assembly (Additional file 2: Fig. S9). Each intron is located at the same position as the homologous intron in other angiosperms. The group II introns were all present in the ancestral seed plant mitogenome [37], whereas the group I intron (in *cox1*) is a homing intron that has spread to hundreds of angiosperm lineages via frequent HGT [38]. Five introns are almost certainly *trans*-spliced based on their being *trans*-spliced in all other angiosperms, with no cases of reversion from *trans*- to *cis*-splicing known [37]. Consistent with this expectation is the far-flung separation of flanking exons for four of these introns (Fig. 1B).

Most of the 20 *cis*-spliced introns are exceptionally large in both *B. yakushimensis* and *B. laxiflora* (Fig. 2, Additional file 1: Table S6). The largest intron, 38.8 kb in size, is 3.4 times larger than the previous record-holder in land plants [39]. For several reasons, these large introns are almost certainly still *cis*-spliced (Additional file 2: Fig. S10 and Additional file 3: B) [37, 40].

(See figure on next page.)

Fig. 1 The *B. yakushimensis* mitogenome. **A** Links among the 21 regions that comprise the genome. Arrows and lines represent regions and links between regions, respectively. Arrow lengths correspond to five size bins of the regions (0–1 kb, 1–10 kb, 10–50 kb, 50–100 kb, and > 100 kb). Resolution of all pathways in this diagram generates the six-chromosome genome map in Fig. S1. **B** Annotation and features of the 21 regions (see also Fig. S18). Gene annotation, AT content, and repeats are shown in the inner tracks. Genes in blue and green are transcribed clockwise and counter-clockwise, respectively. Grey boxes represent *cis*-spliced introns. The colored annotation indicates genes that span two regions of Fig. S1

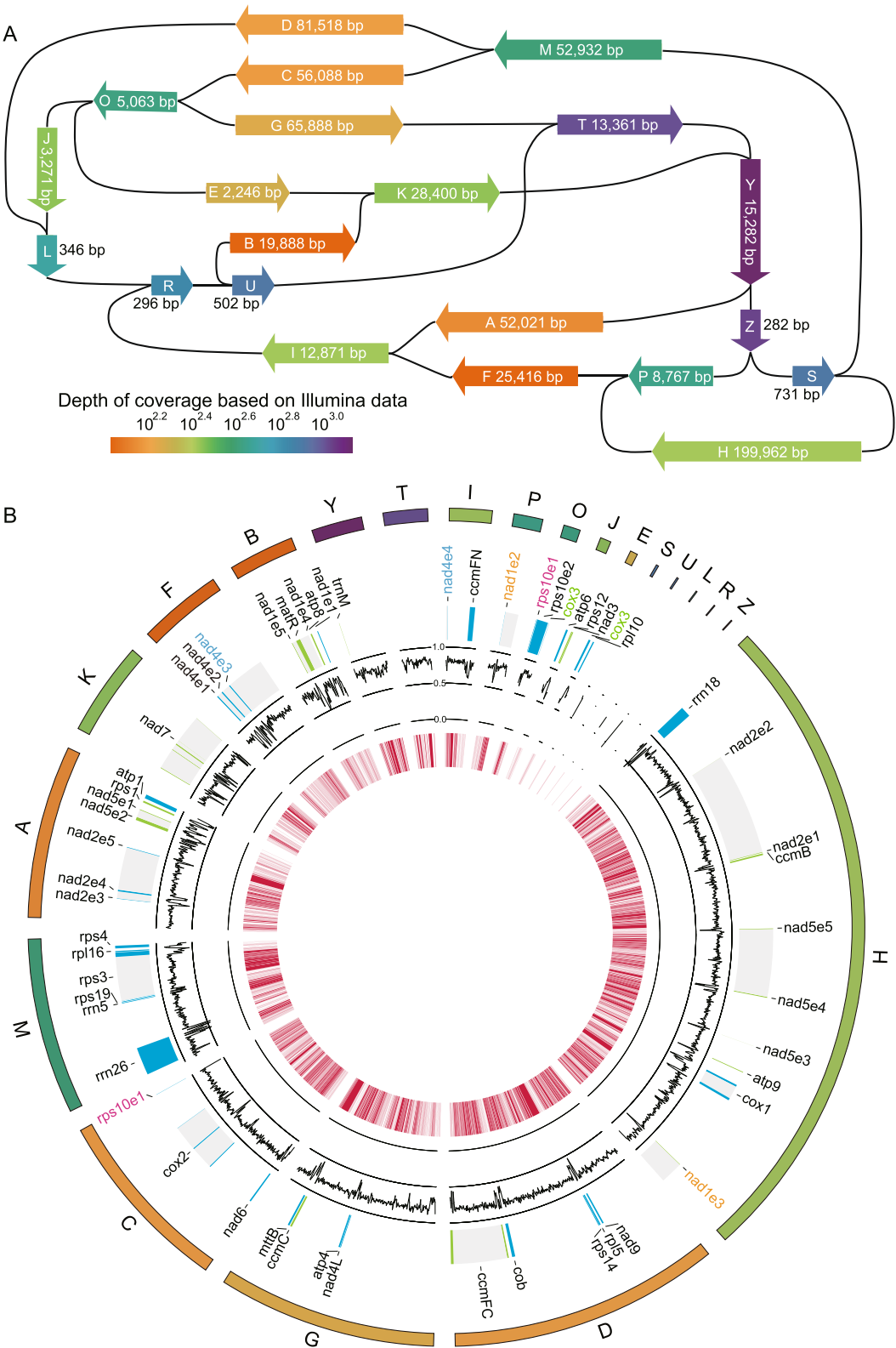


Fig. 1 (See legend on previous page.)

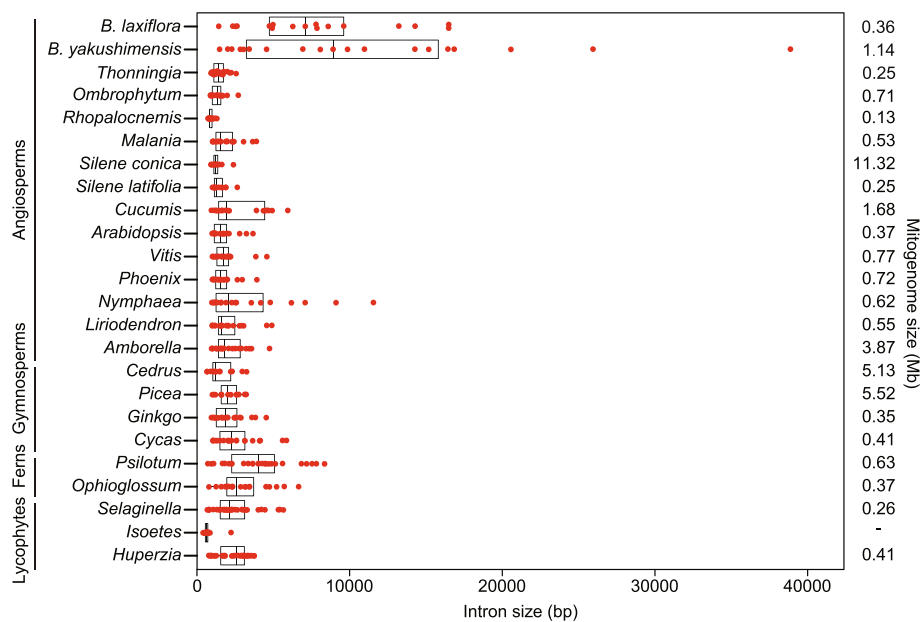


Fig. 2 Variation in *cis*-spliced intron size and genome size in vascular plant mitogenomes. Taxa were chosen to represent phylogenetic diversity and extremes of intron and genome size. Vertical lines internal to boxes, median intron size; boxes, interquartile range (25th to 75th percentile)

Given the enormous size of *Balanophora* introns, it was of interest to determine the location and size of the insertions responsible for their expansion with respect to the conserved secondary structure of group II introns, which consists of six domains (DI–DVI) radiating from a central wheel. We did this for the seven *cis*-spliced introns for which secondary structure models are available in other angiosperms (Additional file 2: Fig. S11) [34, 36, 41–44]. Insertions in DI and DIV account for roughly 99% of the expansion of these seven introns, with this expansion dominated by large insertions (Additional file 1: Table S7). For example, the largest *B. yakushimensis* intron, nad2i1, contains a giant insertion, of 21.7 and 16.1 kb, in DI and DIV, respectively. DIV has a very large insertion (>1 kb) in six of the seven examined introns and five moderately large insertions (100–1000 bp), and DI has a very large insertion (all >2 kb) in four introns and four moderately large insertions (Additional file 1: Table S7). No large insertions were found in the other four intron domains. The comparable incidence of large insertions in DI and DIV of *B. yakushimensis* mitochondrial introns is surprising given that all but one large insertion in published land-plant data are located in DIV (Additional file 1: Table S8) [39, 43, 45–59].

We were also able to examine DV and DVI from all 19 *cis*-spliced group II introns in *B. yakushimensis*. As expected given the highly conserved nature and functional importance of DV [35, 60], eight introns have no insertions (or deletions) in DV, while the total of 12 DV insertions in the other 11 introns are all short, nine of

them just 1–3 bp in length (Additional file 1: Table S6 and Additional file 2: Fig. S11). Indels are more common in DVI and often relatively large, most notably a 128-bp insertion in rps10i1 (Additional file 4: Fig. S12). In conjunction with substitutions, indels have seriously disrupted the functionally critical base pairing between the ends of DVI for three introns (nad7i3, rps3i1, rps10i1) with classical DVI secondary structures in other examined angiosperms. Such degenerative disruptions have previously been described in wheat for several other group II introns (and are shared by *B. yakushimensis* and the other taxa in Additional file 4: Fig. S12) and shown to be associated with atypical splicing pathways in an intron-specific manner [35].

High AT content driven largely by proliferation of AT-rich repeats

At 20.3% GC, *B. yakushimensis* mtDNA is extremely AT rich, as is the *B. laxiflora* draft assembly (25.3%). These mitogenomes are far more AT rich than any other land-plant mitogenomes, including those of other Balanophoraceae genera (Fig. 3, Additional file 4: Fig. S13). Also remarkable is the much higher GC content in protein-coding sequences (37.9% in *B. yakushimensis*) than in the rest of the mitogenome (19.5%) (Figs. 1B and 3, Additional file 1: Table S9). In stark contrast, in the many diverse angiosperms analyzed in Fig. 3 and Additional file 1: Table S9, mitogenomic GC content is higher, albeit usually only slightly, than protein-coding GC. To gain broader perspective, we analyzed 41 other

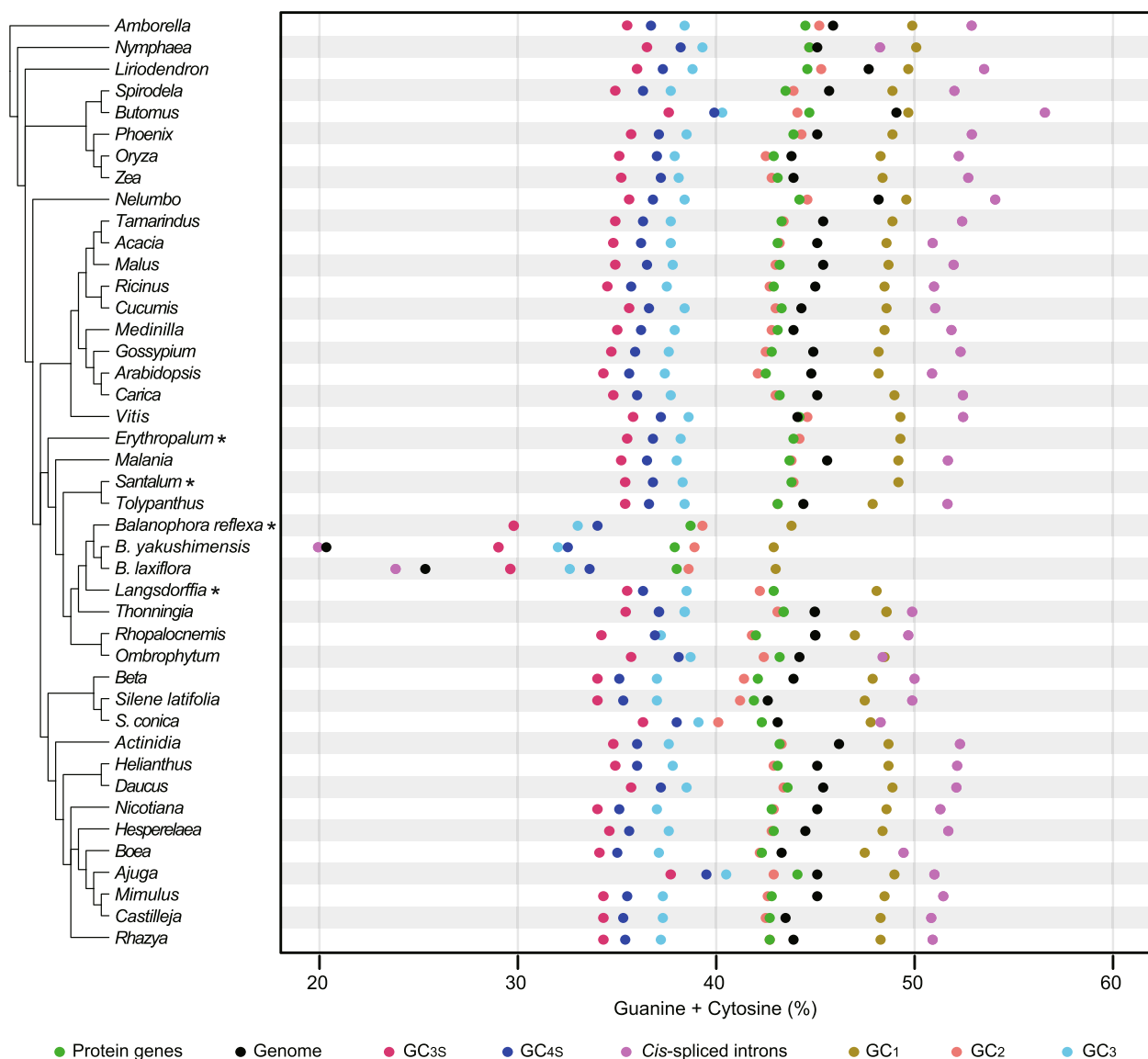


Fig. 3 GC content of 43 angiosperm mtDNAs. Genic values are for 24 core mitochondrial genes, i.e., genes rarely if ever lost from angiosperm mitogenomes. GC3S, GC content at 3rd position synonymous sites; GC4S, GC content at fourfold synonymous sites. Asterisks mark assemblies that are too incomplete for genome and *cis*-intron GC content to be estimated. See Table S9 for exact values

AT-rich genomes from a broad diversity of mitochondria, plastids, and bacteria (Additional file 1: Table S10 and Additional file 4: Fig. S14) [25]. As in angiosperms, most of these AT-rich genomes have similar genomic and protein-coding GC values, with only six having a >4% differential and only one, the *Polytoma uvella* plastome at 15.7%, approaching the *B. yakushimensis* differential of 17.6%.

Two forces, the first likely underlying the second (see Discussion), are responsible for most if not all of the very high AT content of *Balanophora* mitogenomes. The first is an unusually AT-biased point-mutation pressure,

evident from analysis of the GC content of protein-coding sequences. Although protein-coding GC is high in *Balanophora* relative to genomic GC, it is low (37.9–38.7%) compared to protein-coding GC (41.9 to 44.7%) in the 40 other angiosperm mtDNAs (including three other *Balanophoraceae*) examined in Fig. 3 and Additional file 1: Table S9. All five examined partitions of *Balanophora* coding sites show decreased GC too, with the magnitude of the differential between protein-coding GC and GC of each partition much the same in *Balanophora* as in other angiosperms (Fig. 3, Additional file 1: Table S9). In keeping with these findings, codon-usage bias and, to a

lesser extent, amino-acid usage bias are somewhat more pronounced in *Balanophora* than in other angiosperms (Additional file 1: Table S11 and Additional file 4: Figs. S15 and S16). As with other angiosperms, the very close correspondence in *Balanophora* between the effective number of codons (ENC) and the level predicted based on GC3S alone (Additional file 4: Fig. S17) is evidence for neutral causes of its codon-usage bias [61]. On the three terminal *Balanophora* branches, protein genes have an AT mutation bias ranging from 1.5 to 3.9 (Additional file 1: Table S12), and their expected equilibrium GC content under mutation pressure alone (24.0%) is considerably lower than the observed GC content (37.9–38.7%). Therefore, these genes should, depending on the strength of selection against highly AT-rich coding sequences, become more AT rich over time.

The second force driving *Balanophora* mitogenomes to extreme AT-richness is their massive proliferation of AT-rich repeated sequences. The vast majority of the ~47,000 repeat families in *B. yakushimensis* are not only short (see above) but also very AT-rich (Additional file 2: Fig. S2 and Additional file 4: Fig. S18) with an aggregate base composition (excluding repeat-sequence overlaps) of 17.5% GC. Although many repeat families are spread across the mitogenome (Additional file 4: Fig. S19), many directly repeated elements nonetheless have a highly localized/concentrated distribution (Additional file 4: Fig. S20). Furthermore, nearly 84% of the repeats in *B. yakushimensis* mtDNA are tandem repeats (Additional file 1: Table S3). These last two findings, combined with the AT-richness of these repeats, suggest an important role of replication slippage (slipped-strand mispairing) and microhomology-mediated end joining (MMEJ) [62] in the proliferation of these tandem repeats. Indeed, it is conceivable that essentially all of the AT-rich repeats originated as tandem repeats via these processes, with subsequent recombination leading to the dispersion of repeats and, in some cases, a concomitant shift from a tandem/direct arrangement to an inverted one.

These predominantly short and AT-rich repeats occupy the majority (60–90%; median = 77%) of the length of 13 of the 14 largest *cis*-spliced introns in *B. yakushimensis* and a substantial fraction (24–38%) of its six shortest introns (of length 1.3–3.3 kb) (Fig. 4, Additional file 1: Table S6 and Additional file 4: Fig. S18). Moreover, all 22 large insertions inferred for the seven *cis*-spliced group II introns in *B. yakushimensis* for which intron-domain analyses were performed (Additional file 1: Table S7) are highly AT rich (Additional file 2: Fig. S11). The proliferation of AT-rich repeats is thus the driving force responsible for the remarkable inflation of intron size in *Balanophora* mtDNA. In contrast, the ends of these repeat-ridden introns are relatively repeat poor and GC

rich (Fig. 4, Additional file 4: Fig. S18) and presumably contain conserved elements of functional importance for group I and II intron splicing. With the glaring exception of *Balanophora*, introns constitute the most GC-rich component of angiosperm mtDNAs, typically over 50% GC, and only a bit below 50% in other Balanophoraceae genera (Fig. 3, Additional file 1: Table S9) [63]. With an aggregate base composition in *B. yakushimensis* of only 20% GC, introns therefore illustrate most vividly the dramatic effect of AT-rich-repeat proliferation on the evolution of base composition in *Balanophora* mitogenomes.

The rRNA genes of *B. yakushimensis* also contain many AT-rich insertions. The *rrn26* gene contains a giant, 6.5 kb insertion of 18% GC content and 87% repeat content, as well as six other insertions > 50 bp in length (range = 61–437 bp), all of them AT-rich (12–20% GC) (Fig. 4, Additional file 1: Table S13). These result in a gene that is more than three times the size of most other angiosperms, including the other three genera of Balanophoraceae examined. A 319-bp, 12% GC insertion in *rrn5* nearly triples its size as well, while the two largest insertions in *rrn18*, of 15% and 17% GC, add 1163 bp to its length. The *B. yakushimensis* RNAseq data show that its rRNA insertions are not introns. Six protein genes in *B. yakushimensis* contain relatively large (> 50 bp) insertions, most of them AT rich (Additional file 1: Table S13). These 12 insertions range from 51 to 666 bp in length, for an aggregate length of 2049 bp.

The big picture of repeats in the *B. laxiflora* mitogenome is the same as in *B. yakushimensis*, namely, a great abundance of short, clustered AT-rich repeats that comprise most of the mitogenome, including its large introns, as well as a substantial fraction of its rRNA gene space (Additional file 1: Table S13 and Additional file 4: Figs. S21–S23). However, a major difference lurks below this façade of similarity: Fewer than 0.01% of *B. yakushimensis* repeats have detectable sequence homology with *B. laxiflora* repeats (Additional file 1: Table S14 and Additional file 5: Fig. S24). Furthermore, the *B. laxiflora* repeats are on average less AT-rich (26.8% GC) than the *B. yakushimensis* repeats (17.5% GC) (Additional file 2: Fig. S2 vs Additional file 4: Fig. S21, and Additional file 4: Fig. S18 vs. Fig. S23). Thus, *B. yakushimensis* and *B. laxiflora* possess ostensibly different families of repeats.

Despite their current lack of recognizable similarity, many of the repeat families in these two mitogenomes may nonetheless have a common origin, but one that has been obliterated by frequent replication slippage and MMEJ in conjunction with high, AT-biased point-mutation rates in *Balanophora* mitogenomes. Evidence for this hypothesis comes from the AT-rich insertions in mitochondrial exons: Seventeen of the 19 AT-rich insertions > 50 bp in length present in *B. yakushimensis*

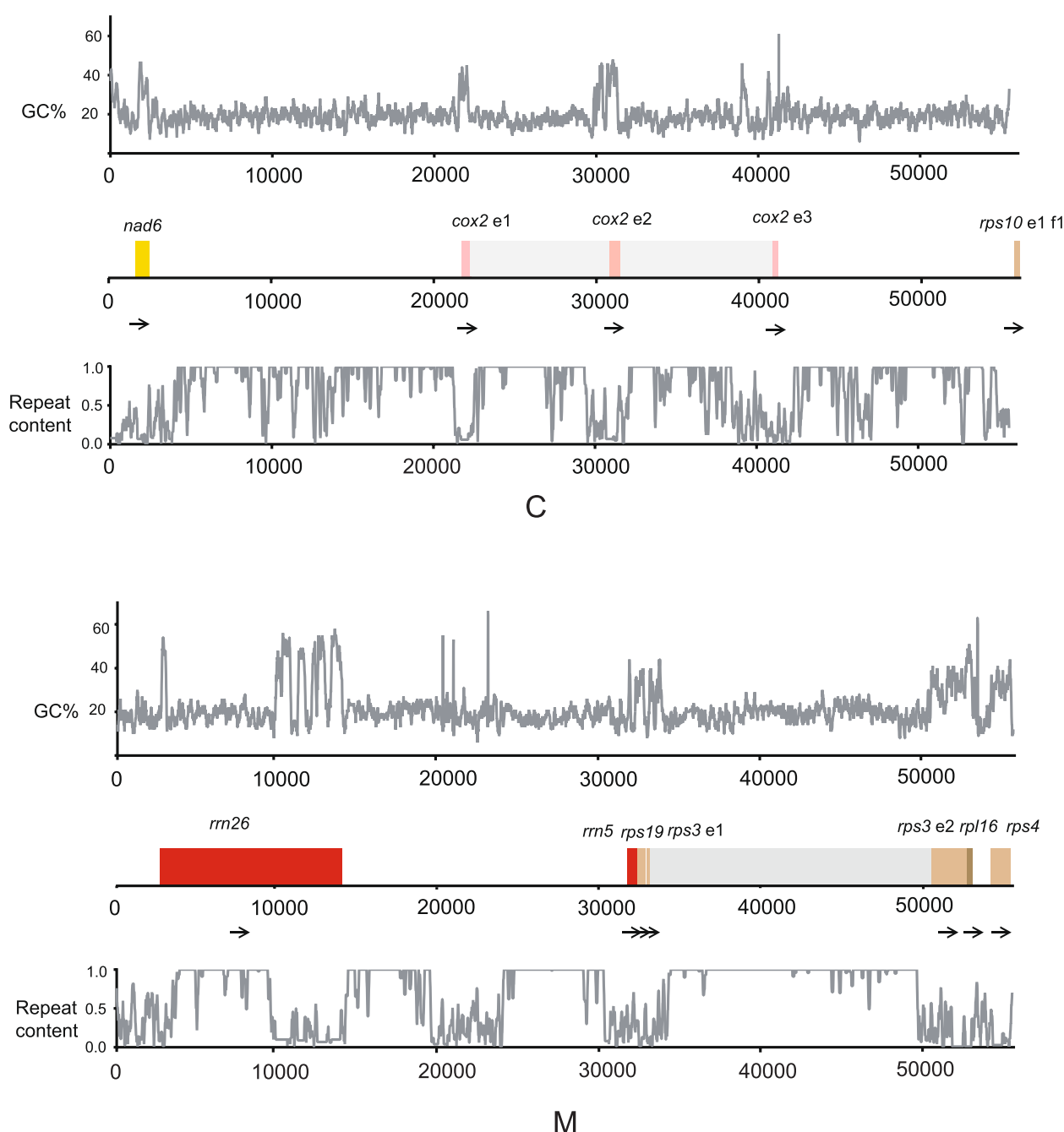


Fig. 4 GC content, repeat content, and gene annotation in regions C and M of the *B. yakushimensis* mitogenome. See Fig. S18 for analysis of all 21 regions

exons are positionally shared with *B. laxiflora*, while at most only one such insertion in *B. laxiflora* is not shared positionally with *B. yakushimensis* (Additional file 1: Table S13). Yet none of these insertions, including those that contain repeated sequences in both mitogenomes, have detectable sequence homology (e.g., Additional file 5: Fig. S25). This is best exemplified by the large (6.5

and 4.8 kb), AT- and repeat-rich, positionally shared insertions in the *rrn26* genes of *B. yakushimensis* and *B. laxiflora*, respectively (Additional file 1: Table S13 and Additional file 5: Fig. S25). In summary, these data support a common origin of many AT-rich repeats in the two *Balanophora* mitogenomes, an origin that was then erased by very rapid divergence of the repeats. This

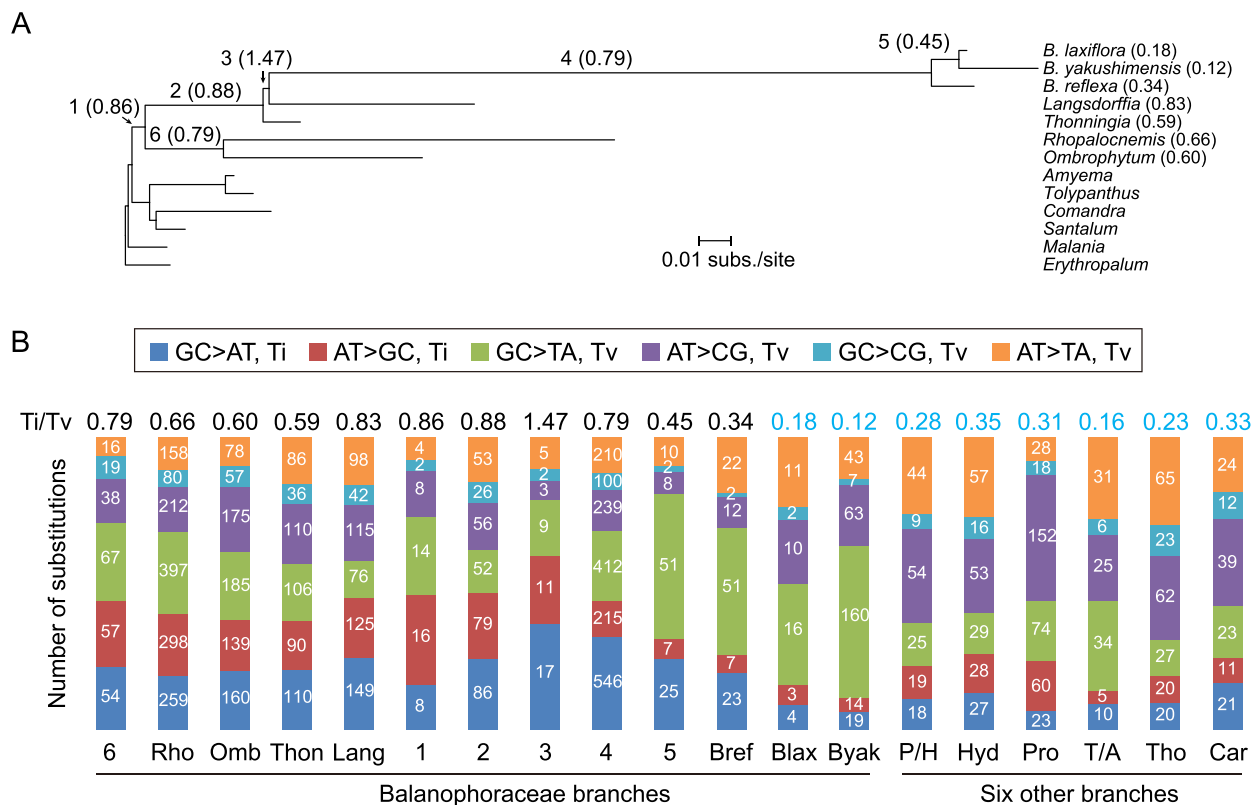


Fig. 5 Variation in mutation spectra. **A** Ti/Tv variation in Balanophoraceae mitochondrial protein genes. Santalales portion of the topologically constrained, synonymous site tree of Fig. S29, with the six internal Balanophoraceae branches numbered. Ti/Tv values are in parentheses. **B** Mutation spectra for Balanophoraceae branches (Table S12) and the six other mtDNA branches with significant transversion biases (Table S15). These six include five Piperaleae branches [P/H, the branch leading to *Hydnora* (Hyd) and *Prosopanche* (Pro); and T/A, the branch leading to *Thottea* (Tho) and *Aristolochia*] and the *Carica* branch (Car). Ti/Tv values of the eight branches with significant transversion biases are in blue

divergence may result in part from the ability of replication slippage to produce substitution errors via transient misalignment [64] and of MMEJ to generate very short (1–4 NT) insertions and deletions [65].

Mutation spectrum evolution

Analysis of a 24,519-bp alignment of 32 mitochondrial protein genes reveals successive decreases in the transition-to-transversion ratio (Ti/Tv) along the branches leading to the three *Balanophora* species (Fig. 5A, Additional file 1: Table S12 and Additional file 5: Fig. S26). These include significant decreases on the terminal *Balanophora* branches, to Ti/Tv=0.12 ($P=1.1 \times 10^{-6}$; chi-square test), 0.18 ($P=0.042$), and 0.34 ($P=8.8 \times 10^{-5}$) for *B. yakushimensis*, *B. laxiflora*, and *B. reflexa*, respectively. There is relatively little change in Ti/Tv elsewhere in Balanophoraceae. The *B. yakushimensis* and *B. laxiflora* branches have similar Ti/Tv values despite differing in length by sevenfold across all sites.

The Ti/Tv values in *B. yakushimensis* and *B. laxiflora* are lower than those reported for any genome,

mitochondrial or otherwise (see Discussion). Therefore, to gain perspective on these exceptionally low values, we calculated Ti/Tv for a concatenate of 24 core mitochondrial protein genes across a topologically constrained tree of 48 diverse angiosperms for which Yu et al. [66] reported mutation spectra, but not Ti/Tv values. For the 59 branches with ≥ 50 substitutions, Ti/Tv ranges from 0.16 to 3.17, with a median of 0.76, and a mean of 0.87 (Additional file 1: Table S15 and Additional file 5: Figs. S27 and S28). Relative to the null of Ti/Tv=0.50 (all substitution types occur at the same rate), 31 branches (including all 29 branches with Ti/Tv ≥ 0.80) have a significant transition bias (two-tailed binomial test), six have a significant transversion bias (Ti/Tv ≤ 0.35 in all six cases), and 22 branches have no significant bias (Additional file 1: Table S15). This analysis identified two more groups of angiosperms within which Ti/Tv has dropped by about an order of magnitude and found many other statistically significant decreases in Ti/Tv, and a number of significant increases, across this tree of 48 angiosperms (Additional file 3: C and Additional file 5: Fig.

S27) [67, 68]. In addition, we corrected the mitochondrial Ti/Tv ratios of Sloan and Wu [68] and found that Ti/Tv ranges from 0.31 to 2.22 (median = 0.90, mean = 0.97) for their 13 taxa with ≥ 50 substitutions (Additional file 1: Table S16 and Additional file 3: C) [68]. Combining these corrected data with those from Additional file 1: Tables S15 and S16 yields a 26-fold range in angiosperm mtDNA Ti/Tv values (0.12–3.17), with a median of 0.79 and mean of 0.86 (Additional file 1: Table S17) [68]. Raising the threshold for number of substitutions from ≥ 50 to ≥ 100 , ≥ 200 , and ≥ 300 has little effect on median and mean Ti/Tv values, while reducing the range by only 23–30% (Additional file 1: Table S17). Thus, our findings of relatively low yet highly variable Ti/Tv values in angiosperm mtDNA are robust to a wide range of substitutional thresholds.

We also examined the full spectrum of substitutions within Balanophoraceae and on the six other branches with significant transversion biases for which mutation-spectrum data are available. Apart from the highly reduced Ti/Tv ratios in *Balanophora* described above, the most notable change in mutation spectrum within the family is a large increase in the proportion of GC > TA transversions (i.e., G-to-T and C-to-A mutations) on the four within-*Balanophora* branches (Fig. 5B). This increase reflects both the Ti/Tv reduction and the reduced GC content of mitochondrial protein genes in the genus (Fig. 3). The three terminal *Balanophora* branches all show an AT mutation bias (1.5–3.9; Additional file 1: Table S12), suggesting that these protein genes will become more AT rich over time.

We found substantial variation in mutation spectra among transversion-biased branches, even among relatively closely related branches, and, conversely, cases of convergence on similar spectra among relatively distantly related branches (Fig. 5B, Additional file 1: Table S18). The most notable case of significant divergence is that of the holoparasite *Prosopanche* (Hydnoraceae) relative to its subtending, P/H branch and its sister genus *Hydnora* (also holoparasitic); this is despite highly similar Ti/Tv ratios on the three branches. Also notable is the divergence of *Thottea* relative to its subtending, T/A branch, with this divergence, in turn, accompanied by convergence between the *Thottea* spectrum and those of *Hydnora* and its subtending branch. Convergence has also occurred between these last three spectra and that of the distantly related *Carica* branch. Finally, the mutation spectra of the T/A branch and *B. laxiflora* have converged.

High rates of nucleotide substitutions

Root-to-tip synonymous substitution (d_s) branch lengths for each of 32 mitochondrial genes are high in the three *Balanophora* species (Fig. 6), and are consistently higher than in all but six other species (*Pelargonium*, *Phoradendron*,

Plantago, *Silene conica*, *S. noctiflora*, and *Viscum*), which have all previously reported to have extremely high rates of synonymous substitution in mtDNA [32, 33, 69–71]. A topologically constrained tree of a concatenate of 23 core mitochondrial genes reveals that root-to-tip d_s values are far longer for the three *Balanophora* species than for all other included taxa (the above six high-rate taxa were excluded to avoid tree-scale compression) and that most of this divergence accrued on the very long branch leading to the *Balanophora* common ancestor (Fig. 5A, Additional file 5: Fig. S29). The nearly tenfold difference in terminal d_s branch lengths for the sister pair *B. yakushimensis* and *B. laxiflora* speaks clearly to further, taxonomically recent rate increase and/or decrease, while there is suggestive evidence for additional rate changes within Balanophoraceae (Additional file 3: D) [7].

Gene-to-gene heterogeneity in root-to-tip substitution rates, while clearly evident in Fig. 6, is actually low in the three *Balanophora* mitogenomes for both d_s and d_N compared to most angiosperms (Additional file 1: Tables S19 and S20). Values of d_N/d_s average 0.32 in all three *Balanophora* species; this is lower than for most angiosperms (Additional file 1: Table S21 and Additional file 5: Fig. S29) and consistent with other evidence that the well-documented extreme range in synonymous substitution rates in angiosperm mtDNAs is driven primarily by underlying changes in the point mutation rate.

Loss of DNA-RRR genes

In plants, all proteins involved in organellar DNA RRR are encoded by the nuclear genome. Mutants for certain DNA-RRR genes exhibit elevated instability or substitution rates in their organellar genomes compared to wild-type [72–82]. We therefore hypothesized that the exceptional characteristics of both organellar genomes of *B. yakushimensis* may result in part from dysfunction of its DNA-RRR system.

We searched the transcriptome and nuclear genome of *B. yakushimensis*, the nuclear genomes of *B. fungosa* and *B. subcupularis* [28], and the nuclear genomes of the Santalalean hemiparasites *Malaria oleifera* (Ximeniaceae) and *Santalum album* (Santalaceae) for 39 genes putatively involved in organellar DNA RRR. We also re-examined the transcriptomes of *Lophophytum*, *Rhopalocnemis*, and the Santalalean hemiparasite *Taxillus chinensis* (Loranthaceae). Furthermore, we searched for those DNA-RRR genes not yet assayed in *Silene noctiflora* [83], the only angiosperm besides Balanophoraceae with a highly aberrant mitogenome sequence for which a transcriptome and/or nuclear genome sequence are available (both in this case).

All 39 DNA-RRR genes were recovered in intact form from *Malaria*, *Santalum*, *Taxillus*, and *S. noctiflora*.

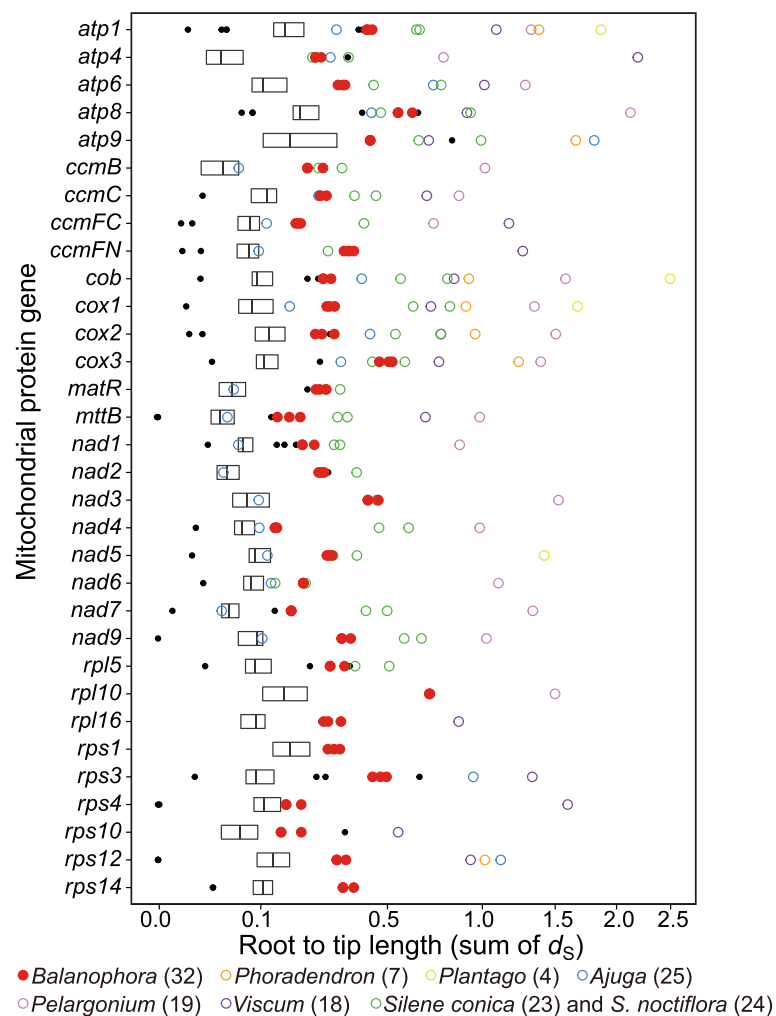


Fig. 6 Root-to-tip synonymous-site (d_s) branch lengths of 32 mitochondrial protein genes from 98 diverse angiosperms. Vertical lines, median values; boxes, interquartile range (25th to 75th percentile); black dots, outliers exclusive of many of the colored circles. Red-filled circles mark values for *Balanophora*. Open circles mark values for taxa with very high d_s values for a minority of genes (*Ajuga*) or a majority (other taxa). The number of genes used for these taxa is in parentheses

In contrast, we failed to recover any portion of 13, 12, 12, eight, and seven DNA-RRR genes from *B. yakushimensis*, *B. fungosa*, *B. subcupularis*, *Rhopalocnemis*, and *Lophophytum*, respectively (Additional file 1: Table S22) [7, 28, 72, 73, 75–78, 83–119], with a total of 14 genes putatively lost on one branch of the family or another. Entirely consistent with these results are those from global gene-family clustering (Additional file 1: Table S23) [28, 119, 120]. Twelve DNA-RRR genes were lost from all three *Balanophora* species, with *FPG1* lost from *B. yakushimensis* only (Additional file 3: E).

Depending on the gene, it is more-or-less unlikely that the 14 DNA-RRR genes apparently missing from one or more *Balanophoraceae* nuclear genomes are actually present, i.e., that we failed to detect them and/

or their transcripts. A false negative result is most unlikely for the six DNA-RRR-gene losses inferred in the *Balanophoraceae* common ancestor (Fig. 7) [7, 8, 20, 23, 121–123] given the congruent results obtained for all five examined *Balanophoraceae* species and for both the transcriptome and nuclear genome of *B. yakushimensis*. Phylogenetic congruence among the three examined *Balanophora* nuclear genomes together with transcriptome/genome congruence for *B. yakushimensis* makes the inferred losses of six additional genes (*OSB1*, *OSB2–4*, *OSBX*, *RECA3*, *RECG1*, and *RECX*) in the common ancestor of *Balanophora* also unlikely to be false negatives. Also unlikely is the inferred loss of *FPG1* in *B. yakushimensis* given this transcription/genome congruence in conjunction with the collinearity

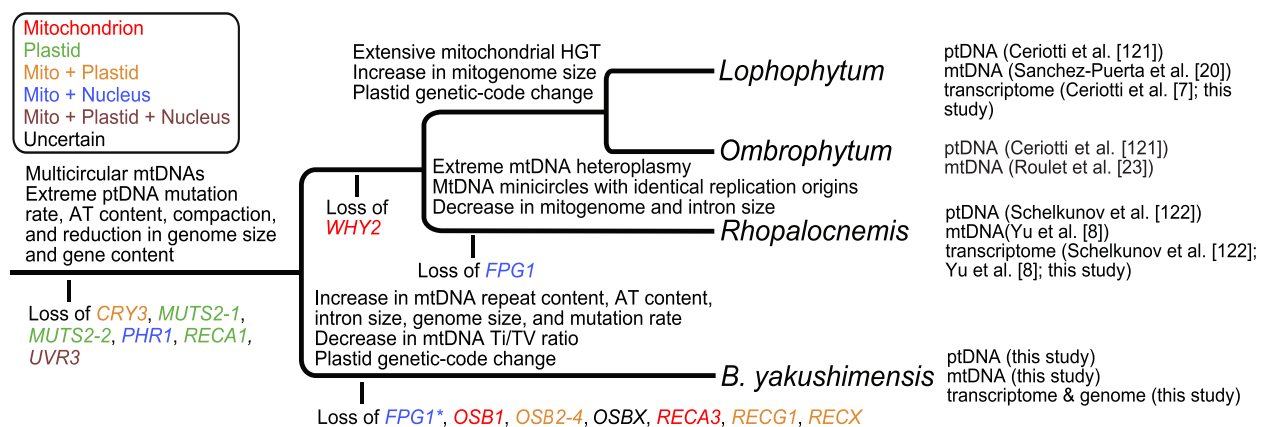


Fig. 7 Loss of organelle DNA-RRR genes and features of organelle genome evolution in Balanophoraceae. Colors denote targeting status as per key at top left. **FPG1* is present in *B. fungosa* and *B. subcupularis*, which otherwise lack all genes that are missing from *B. yakushimensis* and have no additional losses (Table S22). Phylogenetic relationships are based on Ceriotti et al. [121] and Kim et al. [123]

and identity analyses described in Additional file 3: E. The evidence for *Why2* loss is weaker, being based only on transcriptome data, albeit from both *Lophophytum* and *Rhopalocnemis* (Fig. 7). The only apparent loss represented by only a single source of data is that of *FPG1* in *Rhopalocnemis*, and so this is the best candidate for being a false negative. However, we cannot rule out the possibility that any of these genes might nonetheless still be present, but be undetectable as a result of either very low expression (for *Why2* and *FPG1* in *Rhopalocnemis*) or, more generally, of exceptionally rapid sequence evolution. On the other hand, such evolution could in itself be related to an altered or reduced function that contributes to organellar genome aberrancy.

Discussion

Unparalleled repeat content and impact in *B. yakushimensis* mtDNA

Mitochondrial DNA of the holoparasite *B. yakushimensis* possesses an unprecedentedly massive proliferation of short AT-rich repeats that pervade the genome. This, together with its many and outsized large repeats, results in the most repeat- and AT-rich mitogenome known in land plants. The short-repeat proliferation has led to the largest known organellar introns and major exon expansion in rRNA genes and certain protein genes. The *B. yakushimensis* mitogenome also has a record low transition/transversion ratio (0.12), the most AT-rich coding sequences in seed plants, and a highly elevated point-mutation rate. This constellation of extreme features makes this one of the most bizarre and intriguing mitogenomes known.

Repeat content varies widely among seed-plant mtDNAs [124, 125]. *B. yakushimensis* extends this range considerably, to 91% of the genome covered by repeats,

with this value exceeded by only one other organellar genome [126]. No other organellar genome comes close to *B. yakushimensis* in having high coverage values (74% and 69%) for both long (> 1000 bp) and short (< 100 bp) repeats, respectively [124]. *Balanophora yakushimensis* and *B. laxiflora* are the only land plants with highly AT-rich mitochondrial repeats and with repeats whose invasive proliferation has expanded many introns and some exons to such astounding sizes.

As described above and discussed in the next two sections, the *B. yakushimensis* mitogenome surpasses all other organellar genomes with respect to how profoundly it is impacted—inflated (in size), deflated (in GC content), pervaded (in introns and genes)—by repeated sequences. It far surpasses all bacterial genomes and is rivaled by only the largest nuclear genomes, which are packed with highly repetitive transposable elements. Like transposons, the short AT-rich repeats that dominate *Balanophora* mitogenomes should be regarded as selfish DNA elements, but unlike transposons these mitochondrial repeats have proliferated primarily by replication slippage and MMEJ run amok.

Repeat proliferation expanded mitogenome and intron sizes in *Balanophora*

Balanophoraceae mitogenomes vary almost ninefold in size and illustrate multiple forces that drive changes in organellar genome size. Major shrinkage occurred in *Rhopalocnemis* mtDNA (131 kb in size), probably owing to unusually strong deletional pressure that also led to exceptionally small introns. Independent expansions occurred in (1) *B. yakushimensis* (1142 kb) via repeat proliferation and expansion, (2) *Lophophytum* spp. (806–822 kb) via massive HGT from legume host plants [17, 20], and (3) *Ombrophytum* (714 kb) via HGT from

diverse angiosperms, especially Asteraceae host plants [23]. The lack of detectable HGT in *Balanophora*, *Rhopalocnemis*, and (except for one tRNA gene) *Thonningia* adds to the growing evidence that the amount of mitochondrial HGT varies widely among parasitic angiosperms, especially holoparasites [4], and shows that levels of mitochondrial HGT can vary hugely within a family.

In seed-plant mitochondria, variation in intron content and size contributes relatively little to genome-size variation, with *Balanophora* clearly exceptional here. Indeed, the two largest angiosperm mtDNAs, of 6.7 and 11.3 Mb in *Silene*, have relatively small introns, smaller even than those of congeners with far smaller mitogenomes (Fig. 2). In contrast, in most fungi and certain protists, variation in intron number and size accounts for most variation in mitogenome size (e.g. [127, 128]), with size expansion occurring through the insertion of homing endonuclease genes and other ORFs (see [129] for an extreme example).

Balanophoraceae mtDNAs contain the largest and smallest introns known in seed plants and ferns (Fig. 2). The tiny introns in *Rhopalocnemis* range from only 577 to 1145 bp [8]; this probably results from strong deletional forces balanced by selection against loss of functionally important intronic regions. Mitochondrial introns in *Balanophora* are predominantly large but also vary greatly in size, from 1.3 to 39 kb (Fig. 2, Additional file 1: Table S6), reflecting the stochastic acquisition and proliferation of the short AT-rich repeats that pervade the mitogenomes. Thus, very different forces, repeat proliferation vs insertion of homing endonuclease genes and other ORFs, have generated giant mitochondrial introns in *Balanophora* vs. fungi. Mitochondrial intron expansion via repeat proliferation has been reported in two other vascular plants and a green alga, but to a much more limited extent (Fig. 2; [39, 130, 131]).

Two forces have driven *Balanophora* mitogenomes to extreme AT richness

The vast majority of mitogenomes are AT-rich, many of them extremely so, as low as 8% GC (Additional file 1: Table S10; [132, 133]). Until now, land plants stood out in two ways: They almost always possess only modestly AT-rich mtDNAs (mostly 40–46% GC, the lowest at 38%), yet also harbor the most GC-rich mitogenomes known (61–71% GC in the lycophyte genus *Selaginella*) (Additional file 4: Fig. S13; [134]). At only 20.3% GC, the *B. yakushimensis* mitogenome is by far the most AT-rich land-plant mitogenome yet described, in stark contrast to other Balanophoraceae genera with conventional compositions of 44–45% GC.

The AT richness of *Balanophora* mitogenomes is driven by two forces, an unusually AT-biased mutation

pressure and the pervasive proliferation of highly AT-rich short repeats via replication slippage and MMEJ. The biased mutation pressure, felt relatively modestly in *Balanophora* genes, probably underlies the extreme AT-richness (17.5% GC) of the short repeats in *B. yakushimensis*. Unlike genes, the repeats are presumably evolving neutrally and should thus experience the AT mutation pressure equally at all sites. As well, initial, stochastic proliferation of short repeats that happen to lie on the AT-rich end of a probability distribution of repeat base compositions—a sort of founder effect—could contribute to the extreme AT-richness of the *B. yakushimensis* repeats. Consistent with this conjecture, genic GC content is nearly identical in *B. laxiflora* and *B. yakushimensis* (Fig. 3), yet the pervasive short repeats are substantially more GC rich in the former (26.8% GC) than in the latter (17.5%).

The proliferation of such AT-rich repeats has resulted in a record-setting 17.6% differential in base composition between the mitogenome in *B. yakushimensis* (20.3% GC) and its protein-coding sequences (37.9%). In contrast, in most AT-rich genomes, notably including *Balanophora* plastomes, protein-coding sequences have a similar GC content to the genome overall (Additional file 1: Table S10). The most comparable deviant to *B. yakushimensis* mtDNA, the plastome of the green alga *Polytoma uvella* with a 15.7% differential, also features a massive proliferation of short AT-rich repeats that have grossly inflated its intergenic spacers and rRNA genes (relevant data were not reported for *Polytoma* protein genes, while it contains only a single, short intron) [135]. These two genomes thus represent a spectacular case of neutral convergent evolution.

Elevated and variable substitution rates in mtDNA

Only eight Balanophoraceae mitogenomes have been examined, yet multiple and sometimes dramatic increases and probably decreases in rates of synonymous substitutions have occurred within the family. These rate changes are driven by underlying changes in the mutation rate. The accumulated divergence in *Balanophora* is exceeded by only a handful of angiosperm groups among the hundreds examined. These fast groups also harbor multiple rate changes, both increases (some huge in magnitude) and decreases [69–71, 136, 137]. All this, in conjunction with the generally variable synonymous-site branch lengths seen in Additional file 5: Fig. S29 and Mower et al. [138], emphasizes that the point mutation rate is a remarkably fluid property of mtDNA in angiosperms, including parasitic plants. Why this is so, and why most angiosperms nonetheless have very low mutation rates, are long-standing mysteries that are only now being unraveled ([9, 10], see below).

Exceptional Ti/Tv bias in Balanophoraceae and other angiosperms

Previous studies have shown that transitions almost always outnumber transversions, often greatly so, across the tree of life and in all three eukaryotic genomes. This is despite the null hypothesis of $Ti/Tv=0.5$, as there are twice as many types of transversions as transitions. With but one exception [139], all examined bacterial and nuclear genomes have transition biases, with Ti/Tv typically about 1–3 (Additional file 1: Table S24 [140–166]; references 1–18 and 30 in [167]). Vertebrate mtDNAs have famously high transition biases, with Ti/Tv typically about 6–10 [168], while Ti/Tv is typically lower (1–3) in ptDNAs and fungal mtDNAs (e.g. [169, 170]).

Our study considerably extends previous evidence that angiosperm mtDNAs possess relatively low, but variable Ti/Tv values (e.g. [68, 69]), increasing the known cases (in all organisms and genomes) of significant transversion bias from three to 10, uncovering five record-setting Ti/Tv values below 0.30 and as low as 0.12, and extending the range of reported Ti/Tv values to 26 (0.12–3.17). Although most angiosperm mtDNAs possess a transition bias, it is typically quite modest (median Ti/Tv of 0.79), often insignificant, and on-average substantially lower than in all other examined groups.

Why are Ti/Tv ratios so low in angiosperm mtDNA? The most obvious explanation is that this is somehow related to the famously low point-mutation rates in most angiosperm mtDNAs. However, this appears not to be the case. *Balanophora* provides some of the best evidence against this hypothesis, with the sister-pair *B. yakushimensis* and *B. laxiflora* having very similar (and extremely low) Ti/Tv values despite a sevenfold difference in branch length, and with its longest branches unexceptional in Ti/Tv (Fig. 5A, Additional file 1: Table S12 and Additional file 5: Fig. S26). Additional evidence against this hypothesis is presented in Additional file 1: Table S25 and Additional file 3: F [69].

We found substantial variation in overall mutation spectra among the eight mtDNA branches with significant transversion bias. This adds to the growing evidence that mutation spectra can evolve quickly, not just on a phylogenetic scale, but even at the population level [171]. What is novel, to our knowledge, is the discovery of evolutionary convergence in mutation spectra. Finding multiple cases of convergence among only eight selected mitogenomes suggests that this may be a relatively common phenomenon worthy of further investigation.

Contrasting bizarreness in Balanophora mitochondrial and plastid genomes: Vive la différence

Many of the deviancies in the highly divergent mitochondrial and plastid genomes of *Balanophora* run in opposite

directions and reflect radically different deletion/streamlining pressures: The tiny plastome is highly streamlined [25], while the mitogenome is unusually large and spacious. Compaction-related differences between the *B. yakushimensis* mitogenome and plastome involve genome size (1142 kb vs 15 kb, respectively), repeat content (91% vs 5%), intergenic spacer content (65% vs 5%), exon content (3.7% vs 93%), *cis*-intron content (30% vs 0%), *cis*-intron number (20 vs 0), protein-gene number (a normal 34 vs. an ~eightfold reduced 14), and rRNA gene size (much expanded vs normal). A striking expression of these differences is that eight mitochondrial genes and six mitochondrial introns in *B. yakushimensis* are bigger than its entire plastome, with the largest intron nearly three times larger.

The radically different compaction of *B. yakushimensis* organellar genomes is part of a greater puzzle: Why are plastomes, but never mitogenomes, relatively densely packed with genes in seed plants, with even the least compact plastomes more compact than the most compact mitogenomes [7, 33, 172, 173]? These compaction differences presumably reflect a combination of selection, constraints, and more-or-less neutral mutational forces. High plastid gene density could result from selection for compact genomes, a high deletion pressure, the organization of many genes into operons, a general paucity of repeats and of mechanisms for generating repeats, and an inability to acquire foreign sequences via HGT and IGT. Low mitochondrial gene density could reflect opposites in all respects. A clear mechanistic underpinning is available for one of these properties, HGT, whose high rate in seed-plant mtDNA is driven by frequent fusion of mitochondria, whereas plastid fusion is extremely rare [20, 174].

Multiple factors, including loss of certain DNA-RRR genes, contribute to extreme evolution of organellar genomes in Balanophoraceae

Many types of alterations could contribute to the extreme evolution of Balanophoraceae organellar genomes. The most obvious candidates are the 14 organellar DNA-RRR genes lost variously during Balanophoraceae evolution (Fig. 7) and the extreme life-history of all family members as holoparasites living underground during most of their life cycle. Cai [175] proposed that the loss of photosynthesis, the dominant metabolic and biochemical process in photoautotrophs, triggers a suite of population-genetic changes, including relaxed selection and reduction in effective population size, and a cascade of gene losses across plastid and nuclear genomes, some that follow more-or-less directly from the loss of photosynthesis and others relating to the altered population-genetic regime and/or the general establishment of a holoparasitic or

mycoheterotrophic lifestyle. Thus, some gene losses that would be lethal in a photoautotroph are postulated to be permissible in obligate heterotrophs (see [176] for examples). Therefore, while some DNA-RRR gene losses in Balanophoraceae may be causally related to aberrant plastome and/or mitogenome features, others may be only incidentally related, and still others (see below on photolyase genes) may be completely unrelated.

Six DNA-RRR genes were lost in the common ancestor of Balanophoraceae (Fig. 7). The loss of plastid-targeted *RECA1* was almost certainly a crucial event contributing to extreme plastome evolution at the base of the family, indeed to four facets of this evolution, i.e., high substitution rates, high AT content, genome compaction, and genetic-code changes (Additional file 3: G) [4, 6, 25, 73, 75, 177–185]. Two other plastid-specific genes, *MUTS2-1* and *MUTS2-2*, were lost in the Balanophoraceae common ancestor, but there is no evidence on their function in eukaryotes.

The other three DNA-RRR genes lost in the Balanophoraceae ancestor, *CRY3*, *PHR1*, and *UVR3*, encode photolyases, enzymes that repair different types of DNA lesions produced by UV light (Additional file 1: Table S22). These are the only known photolyase genes in angiosperms. We postulate that these losses are unrelated to the highly elevated substitution rates in Balanophoraceae plastomes and instead are a consequence of the common ancestor having lived in deep shade, exposed to so little UV light that selective pressure to maintain photolyase genes was greatly reduced or lost entirely, leading to loss of these genes. This deep-shade hypothesis rests on seven points (Additional file 3: H) [90, 105, 186–188] and predicts frequent loss of photolyase genes, or relaxed selection on them [186], in the many other lineages of deep-shade angiosperm heterotrophs and also in certain lineages of photoautotrophic plants (Additional file 3: H).

Seven DNA-RRR genes were lost on the *B. yakushimensis* branch (Fig. 7). Five of these genes (*OSB1*, *OSB2-4*, *RECA3*, *RECG1*, and *RECX*) are known or thought to be involved in homologous recombination and/or genome-stability in plant mtDNA (Additional file 1: Table S22; [79, 80]), with three of these genes potentially targeted to the plastid as well (see Additional file 3: I) [72, 76, 78, 90, 103, 105, 116, 175, 176, 189, 190]. Therefore, some of these losses may have contributed to the massive proliferation of short repeats and the exceptional number and size of large repeats in *B. yakushimensis* mtDNA. Most of the short repeats are tandem repeats whose proliferation probably occurred by some combination of replication slippage and MMEJ [62]. Accordingly, the loss of *OSB2-4*, which blocks MMEJ [191], may have been important in triggering this proliferation. The loss of *RECA3* and

RECG1 may also have contributed to the elevated substitution rate in *Balanophora* mtDNA as they are important players in recombinational repair (i.e., homologous recombination-dependent DNA repair), arguably the major mechanism of DNA repair in plant mtDNA [80].

As described in Additional file 3: J [81, 88, 191], the independent losses of *FPG1* in *B. yakushimensis* and *Rhopalocnemis* and the loss of *WHY2* in the *Rhopalocnemis* + *Lophophytum* + *Ombrophytum* clade are probably of little consequence. No experimental evidence is available regarding the function or subcellular localization of *OSBX*, absent from *B. yakushimensis*, and localization predictions are inconclusive (Additional file 1: Table S26) [7, 25, 83, 121, 192].

Less drastic mutations than entire gene loss, i.e., point mutations and indels, almost certainly contribute to the aberrant properties of organellar genomes in Balanophoraceae. Intriguing evidence consistent with this comes from *Silene noctiflora* and *S. conica*, whose remarkable mitogenomes are huge (6.7 and 11.3 Mb in size, respectively), with highly elevated rates of point mutations and indels, unusually low rates of repeat-mediated recombination and gene conversion, and numerous circular chromosomes [32]. Yet none of the 39 organellar DNA-RRR genes examined in this study is missing from the *S. noctiflora* nuclear genome, nor are any of the 19 mitochondrial DNA-RRR genes from *S. conica* examined by Havird et al. [83]. Some of this extreme disparity is undoubtedly related to *Silene* being a high-light photoautotroph with relatively normal plastomes [193]. But it is hard to imagine that major-effect point mutations and indels have not occurred in some *Silene* genes involved in mtDNA RRR. Similarly, consider *Rhopalocnemis*, whose exceptional mtDNA features (Fig. 7) are accompanied by only a single DNA-RRR gene loss, this of unlikely relevance (Additional file 3: I). Our data are, however, inconsistent with the occurrence in *Balanophora* of major-effect mutations in *MSH1*, probably the key gene in maintaining low mutation rates in plant organellar genomes [81, 194]. This is because mitochondrial ATGC transitions increase dramatically in an *Arabidopsis* *MSH1* knockout [81] but are uncommonly rare in *Balanophora* (Fig. 5).

Our findings rule out mutagenic retroprocessing, a proposed mechanism for elevated substitution rates in plant mtDNA. The extensive loss of introns and RNA editing supports this mechanism in certain rate-elevated lineages (e.g. [195]). However, *B. yakushimensis* contains a full complement of introns and a nearly typical complement of RNA editing sites.

Properties unrelated directly to canonical DNA-RRR genes, or reverse transcriptase, can contribute to extreme evolution of organellar genomes. Notably, plant mtDNA

copy number was recently shown to underlie ~ 50% of the huge and widespread variation in mitochondrial mutation rates across angiosperms via affecting the efficacy of recombinational repair [10]. These authors propose additional factors that could impinge on the efficacy of this key repair mechanism, including mitochondrial size, mitochondrial fusion/fission dynamics, relaxed selection, and reduction in effective population size.

The chemical environment of genomes can also affect their evolution. Of relevance here, reactive oxygen species (ROS) predominantly cause GCTA transversions [196], which are the major contributor to the very low mitochondrial Ti/Tv ratios in *Balanophora* (Fig. 5). This raises the intriguing possibility that defects in mitochondrial respiration (causing high ROS production), ROS scavenging, and/or repair of ROS-generated DNA lesions arose during *Balanophora* evolution, leading to low mitochondrial Ti/Tv values.

Conclusions

Prior to this study, vascular-plant mtDNAs were known to vary substantially in many ways, including genome size; extent of heteroplasmy; content of genes, introns, RNA edit sites, repeats, plastid-derived DNA, and HGT-derived DNA; and rates of interspecific and intragenomic synonymous substitutions, sequence turnover, rearrangements, repeat-mediated recombination, and gene loss accompanied by functional transfer to the nucleus (for most features, see [3]; for others, see [8, 124, 197]). The *B. yakushimensis* mitogenome adds genomic AT content, intron size, and rRNA gene size to this remarkable panoply of highly variable genomic traits and also substantially extends the range of Ti/Tv ratios, coding-sequence base composition, and percent genome occupied by repeats. Indeed, the constellation of extreme features exhibited by this record-setting genome makes it one of the most bizarre and intriguing mitogenomes known.

Of comparable importance, we found that the nuclear genome of *B. yakushimensis* has lost a record number of genes that function in organellar DNA RRR. We have explored the many ways in which these losses—and other alterations—might or might not be related to the unusual features of both the mitochondrial and plastid genomes of *Balanophora*. Here are two notable outcomes of this exploration: Our mutation-spectra data suggest that unusually high levels of reactive oxygen species in *Balanophora* mitochondria may have contributed to its exceptionally low mitochondrial Ti/Tv ratio. We developed the “deep-shade” hypothesis to account for the loss of three DNA repair genes, all of them photolyase genes, in the common ancestor of the Balanophoraceae.

Our findings should stimulate scientists to examine the content of nuclear genes for mtDNA RRR in the many

exceptional if not outright bizarre mitogenomes found across eukaryotes (including in animals and fungi), for which very few such studies are available [198, 199]. Our study should inspire testing of the deep-shade hypothesis across plants and even, where relevant, non-plants. Our findings make it crucial to thoroughly examine at the sequence level the co-evolution of organellar genes and proteins in Balanophoraceae with their interacting nuclear/organellar partners, as well as to examine its still-resident organellar DNA RRR genes for mutations that could contribute to the many aberrant properties of both organellar genomes in the family. Finally, we hope our study will help inspire expanded genomic investigations into the numerous and diverse lineages of parasitic and also mycoheterotrophic plants.

Methods

Plant samples and sequencing

Plants of *B. yakushimensis*, which were parasitizing a *Berchemia* species, were collected in Shimentai National Forest Park, Yingde, Guangdong, China. The voucher specimen (Yu 20,180,901) was deposited at Sun Yat-sen University Herbarium (SYS). We extracted total DNA of one plant using the Hipure Plant DNA Mini Kit (Magen, Shenzhen, China). We carried out long-read sequencing of a genomic library with 20 kb insert size on a PacBio Sequel II platform; this yielded a total of 51.4 Gb of sequence data (4,107,939 subreads averaging 12,511 bp). In addition, a shotgun genomic library with 350-bp insert size was constructed and sequenced on an Illumina HiSeq X Ten platform, yielding 13.9 Gb of paired-end reads of length 150 bp. We also extracted RNA from male and female inflorescences of *B. yakushimensis* with Trizol reagent (Invitrogen, Carlsbad, CA, USA), obtained mRNA from 1 µg of total RNA from each tissue using polyT magnetic beads, and constructed transcriptome libraries using VAHTS Universal V6 RNAseq Library Prep Kit for Illumina (Vazyme, Nanjing, China) according to the manufacturer's protocol. The RNAseq libraries were sequenced on an Illumina Novaseq platform, yielding 8.1 and 8.5 Gb of paired-end reads of length 150 bp for the male and female tissues, respectively.

Assembly of the *B. yakushimensis* mitogenome

We executed the “correct” function of Canu v1.8 [200] to implement self-correction for the longest 20% of PacBio reads, and further corrected these reads with Illumina short reads using LorDEC v0.6 [201] with default parameters. We used the complete mitogenomes of 15 plants (Additional file 1: Table S27) as seeds to identify putatively mitochondrially derived reads from the corrected PacBio reads, using Blasr v5.3.3 [202] with minAlnLength set to 1000 and minPctSimilarity set to 60. The extracted

reads were fed into Canu to execute “assemble” with the minimum overlap length between reads (minlength) set to 4000 bp and the maximum error rate (maxerate) set to 0.03. This produced 82 contigs with a total length of about 2.7 Mb. To calculate the depth of coverage for these contigs, we mapped the Illumina reads onto these contigs using BWA-mem v0.7.17 [203] with default parameters. We then used the *slim_fastg.py* script in GetOrganelle v1.6.2e [204] to select contigs containing at least one mitochondrial gene. These contigs were used as seeds to identify putatively mitochondrion-derived reads from the corrected PacBio reads with Blasr for a second cycle of assembly, with minAlnLength and minPctSimilarity set to 5000 bp and 95, respectively. After six cycles of assembly with the same Blasr parameters used in the second cycle, 15 contigs remained in the assembly. Ten contigs contained at least one intact mitochondrial gene, had a high AT content of ~80%, and had a relatively consistent depth of coverage of 100× to a few 100 × (Additional file 5: Fig. S30). In sharp contrast, the other five contigs contained only short fragments of mitochondrial genes, had a much lower AT content of 59–64%, and had an extremely uneven depth of coverage (from 0 to >10,000 ×) across the contig. We infer that the 10 AT-rich contigs, which sum to 1,237,083 bp in length, are mitochondrial, whereas the other five contigs are nuclear, containing both single-copy and (sometimes highly) repetitive sequence; the presence of mitochondrial gene fragments in these nuclear contigs is to be expected, as it is well established that angiosperm nuclear genomes have numerous recently transferred and overwhelmingly genetically inert mitochondrial sequences. Pilon v1.3 [205] was used to polish the 10 mitochondrial contigs with the Illumina sequencing reads to correct for sequencing errors in the PacBio reads. To assess the quality of this assembly, we mapped the Illumina reads and the longest 20% of PacBio reads to the 10 contigs using BWA-mem and minimap2 [206], respectively.

Based on self-blast results, the 10 contigs were subdivided at breakpoints of synteny into 21 regions that represent distinct components of the mitogenome (Additional file 1: Table S1). These regions are connected in a complicated manner (Fig. 1A). All links between regions were verified by the PacBio reads with high sequencing depth. Based on Illumina data, the length and depth of coverage of the 21 regions range from 282 to 199,962 bp (sum = 645,144 bp) and from 103 to 1314 ×, respectively, with the PacBio depths of coverage highly similar to the Illumina ones (Additional file 1: Table S1). The depth of coverage is moderately consistent within each region but highly variable among regions, indicating that the regions vary substantially in stoichiometry (Additional file 1: Table S1 and Additional file 5: Fig. S31). Based on the

linkages among these regions (Fig. 1A) and using depth of coverage as an indicator of copy number, we constructed a full-genome assembly of length 1,141,990 bp and consisting of six circular-mapping chromosomes (Additional file 2: Fig. S1; chromosome depth-of-coverage plots are in Additional file 5: Fig. S32). The difference between the aggregate size of the 10 contigs (1,237,083 bp) and the length of the final, full-genome assembly (1,141,990 bp) reflects the presence of redundant overlaps, mostly at their ends, among the contigs. The inferred copy numbers of the 21 regions in the six chromosomes are strongly positively correlated with the observed depth of coverage calculated from both Illumina data ($R^2=0.96$) and PacBio data ($R^2=0.97$) (Additional file 5: Fig. S33). Therefore, the six chromosomes closely reproduce the in vivo stoichiometry of the 21 regions in the *B. yakushimensis* mitogenome. Note that the outliers with respect to this correlation are primarily small regions, for which stochastic effects on depth of coverage, as well as those associated with the abundance of short tandem and dispersed repeats in the mitogenome, should be most pronounced. The sequences of the six chromosomes in this assembly are deposited in GenBank (accession numbers MT892653–MT892658).

Assembly of other genomes

For *B. laxiflora* and *B. reflexa*, plant collection, Illumina sequencing, and mitogenome assembly were performed as described in Su et al. [25] for the plastomes of these species. The 15-contig, 357-kb assembly of *B. laxiflora* mtDNA is described in Results and shown in Additional file 2: Figs. S3 and S4. The *B. reflexa* mitochondrial assembly, of 57 contigs and a total length of only 82 kb, is less complete, but did allow an apparently complete recovery of protein genes (i.e., we recovered the same set of 34 protein genes as from *B. yakushimensis* and *B. laxiflora*). This assembly was used for protein-gene analyses only. For three other Santalales, we used SPAdes v-3.13.0 [207] for mitogenome assembly based on Illumina sequencing data produced by ourselves (*Amyema*, *Comandra*) or downloaded from GenBank (*Santalum*), and then extracted mitochondrial contigs with the script *slim_fastg.py* in GetOrganelle. The above five mitochondrial assemblies are deposited in GenBank (Additional file 1: Table S27).

We assembled the plastome of *B. yakushimensis* using Illumina reads in NOVOPlasty v2.7.1 [208] with default parameters and the *rrn16* gene of *B. laxiflora* as the seed. We carried out PCR amplifications to validate the assembly results. Six overlapping segments covering the whole plastome were amplified using KOD One™ PCR Master Mix (Toyobo, Osaka, Japan) with primers and annealing temperatures listed in Additional file 1: Table S28. Using

the plastid genes of *B. laxiflora* as the reference, we annotated the plastome of *B. yakushimensis* (GenBank accession number MT901289).

We de novo assembled the nuclear genome of *B. yakushimensis* using PacBio long reads and NextDenovo assembler v2.2 (<https://github.com/Nextomics/NextDenovo>) with read_cutoff set to 1 k and seed_cutoff set to 21,370. Nuclear contigs were polished using NextPolish v1.3.0 [209] with PacBio and Illumina reads. To eliminate bacterial contamination, contigs were searched against the NCBI bacterial non-redundant database. Contigs with over 80% of aligned length and over 90% identity to bacterial sequences were discarded. The 4731 retained contigs generated an assembly of 1.24 Gb, with an N50 of 645.1 kb. This assembly has been deposited in GenBank under accession number PRJNA937912.

Gene annotation and expression

To obtain a full set of mitochondrial genes for gene annotation, we made a database consisting of GenBank-available mitochondrial genes of *Balanophora* and 15 other plant species (Additional file 1: Table S27). The 21 regions of *B. yakushimensis*, 15 contigs of *B. laxiflora*, and 57 contigs of *B. reflexa* were used as queries to execute Blastn searches against the database with very sensitive parameters as used in Skippington et al. [210]: word size=7, reward=5, penalty= -4, gapopen=8, and gapextend=6. The boundaries and exon-intron positions of the identified mitochondrial protein genes and rRNA genes were corrected manually as necessary. We used tRNAscan-SE v2.0 [211] with the “organelle” option and default parameters to annotate tRNA genes. The annotated mitogenome maps of *B. yakushimensis* and *B. laxiflora* were created with Circos [212].

To evaluate gene expression and RNA editing, we mapped transcriptome reads from male and female inflorescence tissues of *B. yakushimensis* to its mitochondrial protein genes, using BWA-mem with default parameters. To be conservative, if a mismatch between transcriptome reads and the reference nucleotide site was observed in more than 20% of reads, but there was no mismatch between genomic reads and the same reference site, we considered it an RNA editing site. If fewer than 10 reads mapped to a site, we considered it uninformative.

To determine whether the 25 identified introns were properly spliced, we pooled the RNAseq reads from the two tissues and mapped them onto the coding sequences of the 10 intron-containing genes using BWA-mem with default parameters except with -T set to 130. The RNAseq depth of coverage for the coding regions and the number of read pairs that span splice junctions were calculated with Samtools v1.9 [213].

Sequence alignment of mitochondrial introns and two intron domain regions

To determine the location and size of the insertions that have greatly expanded many *Balanophora* introns, we aligned the entirety of the seven introns (cox2i1, cox2i2, nad1i2, nad1i4, nad2i1, nad5i1, and rps10i1) for which a secondary structure model was available from at least one angiosperm. All seven alignments included *Liriodendron* and *B. yakushimensis*, along with two or three species among *Triticum timopheevi*, *Nicotiana tabacum*, *Vitis vinifera*, *Arabidopsis thaliana*, and *Solanum tuberosum*. We used the published secondary structure models to identify in these alignments the six conserved, secondary structure domains (DI-DVI) of group II introns. We also aligned the terminus, comprising the DV-DVI regions, of all 19 cis-spliced group II introns present in *B. yakushimensis*. These alignments were feasible because DV is short (typically only 34 bp) and highly conserved in size and sequence, with DVI relatively short and sandwiched between DV and the 3' flanking exon, and because secondary structure models of both domains are available for all group II introns of *Triticum timopheevi* [35]. These models, plus those for the two relevant introns absent from wheat (Additional file 1: Table S6), were used to identify DV and DVI in this set of 19 alignments. These alignments had the same taxon sampling as above. All alignments were carried out in MAFFT [214] using the FFT-NS-1 method, followed by manual adjustments.

Mitochondrial repeats and repeat-mediated recombination

We used two approaches to identify repeated sequences (exclusive of the 13 repeated regions in the *B. yakushimensis* assembly) in *B. yakushimensis* and *B. laxiflora* mtDNAs. We used the Python script ROUSFinder.py [215] with the parameter -m set to 30 to identify perfect repeats ≥ 30 bp in size. We also used Tandem Repeat Finder [216] with default parameters (except for a minimum alignment score of 20) to identify tandem repeats, be they perfect or imperfect.

We evaluated repeat-mediated recombination in *B. yakushimensis* at both repeated regions and other repeats. Owing to read-length limitations, potential recombination events mediated by the largest repeated regions (>40 kb) could not be identified. The reference sequence of the assumed recombination for each region-repeat pair was prepared, including the region itself with 1 kb of upstream and downstream flanking sequences. Blasr was then executed to map the longest 20% PacBio reads to the reference sequence with minAlnLength set to the length of the reference sequence and minPctSimilarity set to 95. For each region-repeat pair, if any recombined arrangement was covered by PacBio reads, we

considered that it has recombination activity. For non-region repeats > 500 bp in size identified by ROUSFinder.py, the reference sequence of the assumed recombination contains the repeat itself flanked by 3 or 5 kb of upstream and downstream sequences, respectively, to avoid the influence of nearby repeats.

Searching for IGT

We searched for plastid-derived sequences in *B. yakushimensis* mtDNA in two ways. We searched the *B. yakushimensis* plastome against its mitogenome using Blastn with the parameter -perc_identity set to 80. We also performed Blastx searches using a custom angiosperm plastome database that included *Balanophora* plastomes. We searched for nuclear-derived sequences using RepeatMasker v4.1 [217] with default parameters to identify any transposable elements of nuclear origin in *B. yakushimensis* mtDNA.

Phylogenetic analysis

Coding sequences of 32 of the 34 mitochondrial protein genes present in *Balanophora* (we excluded the two shortest genes, *nad4L* at 264 bp and *rps19* at 258 bp) were collected from 91 diverse angiosperms, including the three species of *Balanophora*, the four previously sequenced Balanophoraceae, and six species from other families of Santalales (Additional file 1: Table S27). Known foreign genes (e.g., from *Ombrophytum*) were excluded, as were short regions in *atp4* and *cox1* that are suspected or known, respectively, to be derived by HGT [23, 38]. Sequences were aligned with MUSCLE v3.8.31 [218], and the alignments were modified manually. Poorly aligned regions were trimmed using Gblocks 0.91b in codon mode [219]. All first- and second-codon-position RNA editing sites listed in García et al. [22] were removed from the alignments to avoid artifacts in phylogenetic inference resulting from the frequent and often substantial parallel loss of RNA editing sites in unrelated lineages [220, 221]. Gene trees were constructed using RAxML [222] with the GTR + Gamma model. One thousand bootstrap replications were performed in all cases. *Amborella* was used as the outgroup for all mitochondrial gene trees except for *rpl10*, for which *Nymphaea* was used. The amino-acid trees in Additional file 5: Fig. S34 [7, 8, 83] of selected nuclear-encoded organellar DNA-RRR genes were constructed as in Ceriotti et al. [7].

Analysis of codon usage

GC content in protein genes, relative synonymous codon usage, and effective number of codons (ENC) of the set of 24 core protein genes from mitogenomes of diverse angiosperms (Additional file 1: Tables S9 and S11), as well as ENC of each mitochondrial protein gene from

B. yakushimensis, were calculated using CodonW v1.4.4 [223]. The expected value for ENC based only on nucleotide composition was calculated using the method of Sloan and Taylor [63].

Estimation of nucleotide substitution rates and mutation spectrum

We used two complementary approaches to analyze mtDNA substitution rates in a large number of diverse angiosperms. In the first, “single-gene” approach, taxa were chosen to include all angiosperms reported or shown in published trees to have long branches, together with a broad phylogenetic diversity of angiosperms with shorter branches. This approach used a codon-based nucleotide model (Codeml in PAML; [224]) to calculate root-to-tip synonymous substitution rates (d_S) for each of 32 genes from 98 angiosperms (Additional file 1: Table S27). The tree topology used for rate estimation was constrained according to the APG system [225], except for some slight modifications [226, 227]. We used the free-ratio model to calculate root-to-tip branch lengths in the d_S and d_N trees for each gene and species to enable comparisons on the same time scale. Codon frequencies were computed using the F1 × 4MG model.

In the second, “concatenate” approach, sequences of 23 “core” mitochondrial genes, i.e., genes rarely if ever lost from angiosperm mitogenomes ([228]; these are the top 23 genes in Fig. 6) were concatenated to estimate aggregate d_S and nonsynonymous (d_N) substitution rates across a topologically constrained maximum-likelihood tree of 92 angiosperms. Sampling here was the same as in the single-gene analyses except that, to avoid several fold tree compression, we excluded the six species reported (also see Fig. 6) to consistently have very high rates of synonymous mitochondrial substitutions: *Phoradendron*, *Plantago*, *Silene conica*, *S. noctiflora*, *Viscum*, and *Pelargonium* (Additional file 1: Table S27). We used Hyphy [229] for this analysis, under a local MG94-F3 × 4-GTR model, because Hyphy is much faster than PAML for a large dataset. In both this and the above rate analysis, *Amborella* was used to root the angiosperm tree, in which case its branch length could not be calculated and therefore it was excluded from all rate figures and tables.

The concatenate approach enabled the single-page display of d_S and d_N values across all tree branches (Additional file 5: Fig. S29), while the single-gene approach enabled the single-page display of d_S and d_N values for each gene separately but only for root-to-tip branches (Fig. 6).

Ancestral sequence reconstruction of 32 protein genes from 97 angiosperms was performed using baseml in PAML, based on the above alignments and tree topology. We counted the numbers of the six types of nucleotide

substitutions along the branches shown in Fig. 5A and Additional file 5: Fig. S26. The expected equilibrium GC content of protein genes was calculated following the method of Lynch [230].

Characterization of DNA-RRR genes

The transcriptome reads from male and female inflorescences of *B. yakushimensis* were pooled and assembled using Trinity v2.8.6 [231] with default parameters except for `-min_kmer_cov` set to 2. TransDecoder v5.5.0 [232] was used to find open reading frames for each transcript sequence. To reduce redundancy, the translated protein sequences were clustered with CD-Hit v4.6.1 [233] with default parameters except for `-c` set to 0.95. Bacterial contamination was removed as above, with transcripts of over 80% of aligned length and over 90% identity to bacterial sequences discarded. To assess the completeness of the transcriptome assembly, the Benchmarking Universal Single-Copy Ortholog (BUSCO) assessment tool ver. 3.0.2 [234] was executed with the "protein" mode, based on the eukaryota_odb10 (255 genes), embryophyta_odb10 (1614 genes), and eudicots_odb10 (2326 genes) datasets. The BUSCO assessment showed that the transcriptome assembly from *B. yakushimensis* has full-length coverage of 94.1%, 57.5% and 53.7% of the conserved, typically single-copy protein genes of the eukaryote, land plant, and eudicot datasets, respectively (Additional file 1: Table S29) [8, 22]. The high recovery of eukaryote genes in *B. yakushimensis* indicates that its transcriptome assembly is more-or-less complete. The major loss of land plant and eudicot genes is entirely to be expected given its holoparasitic lifestyle, with the attendant loss of photosynthesis and a radically simplified morphology, and the similar findings reported in studies on other Balanophoraceae species. These include not only the transcriptome studies on *Rhopalocnemis* and *Lophophytum* (Additional file 1: Table S29) but also the nuclear genome studies on *B. fungosa* and *B. subcupularis* [28]. Indeed, only 51.1–51.3% BUSCO recovery of conserved land-plant genes was reported for these two *Balanophora* species, less than the 57.5% recovery for *B. yakushimensis* land-plant genes in our study; note, however that these values are not strictly comparable because a different number of genes (1375 vs. 1614, respectively) was analyzed in the two studies.

We conducted tblastn searches ($e\text{-value}=1e-3$) against the transcriptome assembly of *B. yakushimensis* using as queries the protein sequences of 39 nuclear *Arabidopsis* genes encoding organellar DNA-RRR proteins (Additional file 1: Table S22). We also did tblastn searches ($e\text{-value}=1e-3$) against the nuclear assembly of *B. yakushimensis* using the same 39 queries. Exon–intron boundaries were determined by comparing genome and

transcriptome sequences. We used the Hmmscan tool from Hmmer v3.3 (<http://www.hmm.org/>) to identify domains in the DNA-RRR genes found in the transcriptome and nuclear genome of *B. yakushimensis*, based on the Pfam-A database. The integrity of domain(s) for each gene was assessed by comparison with the domain information for *Arabidopsis* in TAIR10. We also performed tblastn searches ($e\text{-value}=1e-3$) of the 39 DNA-RRR genes against the nuclear genomes of two other *Balanophora* species, *B. fungosa* and *B. subcupularis* [28], as well as blastp searches ($e\text{-value}=1e-3$) of these genes against the annotated proteins of these two species. Finally, we conducted tblastn searches ($e\text{-value}=1e-3$) of the 39 DNA-RRR genes against the nuclear genome and transcriptome of *Silene noctiflora* [235, 236], the nuclear genomes of *Malania oleifera* [115] and *Santalum album* [119], and the transcriptomes of *Lophophytum mirabile* [22], *Rhopalocnemis phalloides* [8], and *Taxillus chinensis* (Nuclear RefSeq BioProject PRJNA301186 and PRJNA548287). Our re-examination of the *Lophophytum*, *Rhopalocnemis*, and *Taxillus* transcriptomes led to five reassignments of published gene absent/present calls [6–8], all from absent to present.

We also used the AssemblyPostProcessor and GeneFamilyClassifier tools of PlantTribes2 [120] to classify putative DNA-RRR genes of all nine examined species into pre-computed orthogroups of the built-in 22G v1.1 dataset in PlantTribes2. For the two cases in which two DNA-RRR gene models are in the same orthogroup (*OSB1* and *OSBX*; *RECA2* and *RECA3*), we aligned the sequences in each orthogroup in the GeneFamilyAligner tool and constructed the maximum likelihood tree in the GeneFamilyPhylogenyBuilder tool of PlantTribes2 to determine the orthologous sequences for each gene model.

Abbreviations

BUSCO	Benchmarking universal single-copy ortholog
DNA RRR	DNA replication, repair, and recombination
ENC	Effective number of codons
HGT	Horizontal gene transfer
IGT	Intracellular gene transfer
mtDNA	Mitochondrial DNA
MMEJ	Microhomology-mediated end joining
ROS	Reactive oxygen species
Ti/Tv	Transition-to-transversion ratio

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12915-025-02449-8>.

Additional file 1: Tables S1–S29. Table S1 Features of the 21 regions in the mitogenome of *Balanophora yakushimensis*, sorted by Illumina depth of coverage. Table S2 Features of mitochondrial genes of *Balanophora yakushimensis*. Table S3 Repeat sequences in the *Balanophora yakushimensis* mitogenome. Table S4 RNA editing in male and female inflorescences of *Balanophora yakushimensis* for the 29 mitochondrial protein genes with sufficient transcript read depth. Table S5 Diversity of Balanophoraceae

mitogenomes. Table S6 Cis-spliced mitochondrial introns of *Balanophora yakushimensis* and *B. laxiflora*. Table S7 Domain location and estimated size of insertions > 100 bp in length in the seven mitochondrial group II introns of *Balanophora yakushimensis* for which secondary-structure models are available in other angiosperms. Table S8 Length variation in the six domain regions of plant mitochondrial group II introns for which multiple secondary structures are available. Table S9 GC content and effective number of codons (ENC) in angiosperm mitochondrial genes (the 24 core genes) and genomes. Table S10 GC content of 44 AT-rich mitochondrial, plastid, and bacterial genomes. Table S11 Relative synonymous codon usage for the 24 core mitochondrial genes of 41 angiosperms. Table S12 Variation in mitochondrial mutation spectra in Balanophoraceae and a Santalalean outgroup branch based on a 24,519-bp alignment from a concatenate of 32 protein genes. Table S13 Insertions larger than 50 bp in mitochondrial exons of *Balanophora yakushimensis* and *B. laxiflora*. Table S14 All of the very few Blast hits between "non-region" repeat sequences of *Balanophora yakushimensis* mtDNA (with some 47,000 such repeats; Table S3) and *B. laxiflora* mtDNA. Table S15 Ti/Tv ratios across diverse angiosperms based on a 19,309-bp alignment (from Yu et al. [66]) of a concatenate of 24 core mitochondrial protein genes and a constrained phylogeny (Fig. S28) of 48 species. Table S16 Ti/Tv ratios in mtpsts (mtDNA sequences acquired from ptDNA) and 24 core mitochondrial protein genes where available for comparison to mtpst values. Table S17 Ti/Tv variation in angiosperm mtDNA at four thresholds of total number of substitutions. Table S18 Chi-square significance testing (<https://www.icalcu.com/stat/chisqtest.html>) of five sets of nucleotide substitutions among the eight mtDNAs with significant transversion biases (Fig. 5B) as per the two-tailed exact binomial test ($R, 4.3.1$). Table S19 Root-to-tip dS values for 32 mitochondrial protein genes from 96 angiosperms. Table S20 Root-to-tip dN values for 32 mitochondrial protein genes from 96 angiosperms. Table S21 Root-to-tip dN/dS values for 32 mitochondrial protein genes from 96 angiosperms. Table S22 Recovery of 39 putative organellar DNA-RRR genes from the transcriptome and/or nuclear genome of *Balanophora yakushimensis* and eight relevant species. Table S23 Identification of putative organellar DNA-RRR genes from the transcriptome or annotated protein sequences of *Balanophora yakushimensis* and eight relevant species using PlantTribes2. Table S24 Ti/Tv ratios from 34 mutation accumulation (MA) lines and four parent-offspring trio datasets. Table S25 Evidence against a relationship between low Ti/Tv values and low mutation rates in angiosperm mtDNAs. Table S26 Targeting of 40 organellar DNA-RRR genes in *Arabidopsis* for which there is experimental evidence on targeting in this species and also of *OSBX*, which is absent from *Arabidopsis* and for which there is no such experimental evidence. Table S27 Angiosperm mitochondrial sequence data used in this study. Table S28 PCR primers used for validating the plastome assembly of *Balanophora yakushimensis*. Table S29 BUSCO analysis of the completeness of the transcriptome assemblies of *Balanophora yakushimensis*, *Rhopalocnemis phalloides*, and *Lophophytum mirabile*.

Additional file 2: Figures S1–S11. Fig. S1 The six-chromosome model of the *B. yakushimensis* mitochondrial genome generated from the assembly diagram shown in Fig. 1A. Fig. S2 Pairwise plots of AT content, length, and copy number of repeat families in the *B. yakushimensis* mitogenome. Fig. S3 Annotation and GC content of the 15 contigs that constitute the 357-kb draft assembly of the *B. laxiflora* mitogenome. Fig. S4 Depth of Illumina-read coverage of the 15 contigs that comprise the draft assembly of the *B. laxiflora* mitogenome. Fig. S5 Annotation and GC content of the plastome of *B. yakushimensis*. Fig. S6 PCR validation of the Illumina-based assembly of the *B. yakushimensis* plastome. Fig. S7 Gene content (excluding pseudogenes) of the mitogenomes of six Balanophoraceae species. Fig. S8 Maximum-likelihood analysis of protein-coding sequences from 32 mitochondrial genes in as many as 91 diverse angiosperms, depending on the number of species for which a given gene is missing from the species' genomes or their assemblies. Fig. S9. Depth of RNA-seq coverage of the 10 intron-containing mitochondrial genes in *B. yakushimensis*. Fig. S10 Blocks of synteny between the mitogenomes of *B. yakushimensis* and *B. laxiflora*. Fig. S11 Alignment and domain annotation for *B. yakushimensis* and selected other angiosperms for the seven *cis*-spliced group II introns present in *B. yakushimensis* for which secondary structure models are available from other angiosperms.

Additional file 3: Supplementary Results and Discussion. A–No evidence for IGT in *B. yakushimensis* mtDNA or for HGT of protein genes in any *Balanophora* mtDNA or in *Rhopalocnemis* and *Thonningia* mtDNA. B–Large introns in *Balanophora* are still *cis*-spliced. C–Variation in mitochondrial Ti/Tv. D–Variable substitution rates in Balanophoraceae mtDNAs. E–*FPG1* is missing from one of three examined *Balanophora* species. F–No correlation between low Ti/Tv values and low mutation rates in angiosperm mtDNAs. G–The critical loss of *RECA1* in the common ancestor of Balanophoraceae. H–Evidence for the deep-shade hypothesis for loss of all three photolyase genes in the common ancestor of Balanophoraceae. I–Loss of three potentially dual-targeted genes in *B. yakushimensis* only. J–Loss of *FPG1* and *WHY2*.

Additional file 4: Figures S12–S23. Fig. S12 Alignments of DV (yellow), DVI (blue), and associated linker/wheel regions (green) from the 19 *cis*-spliced group II introns present in *Balanophora yakushimensis*. Fig. S13 Variation in mtDNA GC content among 41 angiosperms (A) and 256 land plants (B). Fig. S14 GC content of 44 AT-rich mitochondrial, plastid, and bacterial genomes. Fig. S15 Codon usage in the mitogenomes of four Balanophoraceae and three other angiosperms. Fig. S16 Amino acid usage in the mitogenomes of four Balanophoraceae and three other angiosperms. Fig. S17 Plots of effective number of codons (ENC). Fig. S18 GC content, repeat content and gene annotation in the 21 regions of the *B. yakushimensis* mitogenome. Fig. S19. Self & non-self dot plots of the six largest regions in the *B. yakushimensis* mitogenome. Fig. S20 Number of repeat pairs in *B. yakushimensis* mtDNA as a function of the distance between repeat-pair members. Fig. S21 Pairwise plots of AT content, length, and copy number of repeat families in *B. laxiflora* mtDNA. Fig. S22 Self & non-self dot plots of the eight largest contigs of the *B. laxiflora* mitogenome. Fig. S23 GC content, repeat content, and gene annotation in three contigs (ctg1, ctg2 and ctg10) of the *B. laxiflora* mitogenome.

Additional file 5: Figures S24–S34. Fig. S24 Self and non-self dot plots of five mitochondrial introns from *B. yakushimensis* and *B. laxiflora*. Fig. S25 Synteny plots of mitochondrial rRNA genes from *B. yakushimensis* and *B. laxiflora*. Fig. S26 Low transition/transversion (Ti/Tv) ratios in *Balanophora* mtDNAs. Fig. S27. Widespread variation in mitochondrial Ti/Tv values. Fig. S28 Constrained maximum-likelihood phylogram of 48 angiosperms. Fig. S29 Topologically constrained maximum-likelihood trees based on dS and dN sites from a concatenate of 23 core mitochondrial genes. Fig. S30 Depth of coverage of the 10 assembled contigs of the *B. yakushimensis* mitogenome based on both PacBio and Illumina sequencing data. Fig. S31 Depth of coverage of the 21 regions of the *B. yakushimensis* mitogenome based on both PacBio and Illumina sequencing data. Fig. S32 Depth of coverage of PacBio reads of the six circular-mapping chromosomes that constitute the *B. yakushimensis* mitogenome. Fig. S33 Correlation between copy number and depth of coverage of the 21 regions of the *B. yakushimensis* mitogenome. Fig. S34 Maximum likelihood trees of DNA-RRR genes that have experienced recent duplication events in the lineage leading to *Arabidopsis*.

Acknowledgements

We thank Yuanqiu Li for help with plant sampling, Shuaixi Zhou for drawing the dot-plot figures, and Dan Sloan and José Gualberto for helpful discussion.

Authors' contributions

CWd, JPM, MVS-P, JDP, and RZ designed research; RY, XZ, LFC, ES, DWR, H-J Su, TJB, CS, and YL performed research; RY, XZ, LFC, ES, DWR, CS, YL, DF, XC, JPM, MVS-P, JDP, and RZ analyzed data; and RY, LFC, JPM, MVS-P, JDP, and RZ wrote the paper. All authors read and approved the final manuscript.

Funding

This study was supported financially by the National Natural Science Foundation of China (31811530297 and 32170217) (to R.Z.).

Data availability

The mitogenome data generated in this study have been deposited in the NCBI GenBank (<https://www.ncbi.nlm.nih.gov/>) under accession numbers MT892653–MT892658 (*B. yakushimensis*), ON638215–ON638229 (*B. laxiflora*),

ON638230-ON638286 (*B. reflexa*), MT895602-MT895629 (*Comandra umbellata*), MT895574-MT895601 (*Amyema beccarii*) and ON729993-ON730005 (*Santalum album*). The plastome data of *B. yakushimensis* have been deposited in the NCBI GenBank under accession number MT901289. The nuclear genome assembly data of *B. yakushimensis* have been submitted to the NCBI BioProject database (<https://www.ncbi.nlm.nih.gov/bioproject/>) under accession number PRJNA937912.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹School of Life Sciences, State Key Laboratory of Biocontrol and Guangdong Provincial Key Laboratory of Plant Stress Biology, Sun Yat-Sen University, Guangzhou 510275, China. ²IBAM, Facultad de Ciencias Agrarias & Facultad de Ciencias Exactas y Naturales, Universidad Nacional de Cuyo, CONICET, Chacras de Coria, Almirante Brown 500, Mendoza M5528AHB, Argentina. ³Department of Biology, Indiana University, Bloomington, IN 47405, USA. ⁴Department of Earth and Life Sciences, University of Taipei, Taipei 100, Taiwan. ⁵Department of Biological Sciences, Western Michigan University, Kalamazoo, MI 49008, USA. ⁶State Key Laboratory of Agricultural Genomics, BGI-Shenzhen, Shenzhen 518083, China. ⁷Department of Biology, Pennsylvania State University, University Park, PA 16802, USA. ⁸Center for Plant Science Innovation and Department of Agronomy and Horticulture, University of Nebraska, Lincoln, NE 68588, USA. ⁹Present Address: State Key Laboratory of Plant Diversity and Specialty Crops, Institute of Botany, The Chinese Academy of Sciences, Beijing 100093, China. ¹⁰Department of Bioinformatics and Computational Biology, Genentech, South San Francisco, CA 94080, USA.

Received: 18 July 2025 Accepted: 21 October 2025

Published online: 18 November 2025

References

- Mower JP, Sloan DB, Alverson AJ. Plant mitochondrial genome diversity: the genomics revolution. In: Wendel JH, editor. *Plant Genome Diversity Volume 1: plant genomes, their residents, and their evolutionary dynamics*. New York: Springer; 2012. p. 123–44.
- Smith DR, Keeling PJ. Mitochondrial and plastid genome architecture: reoccurring themes, but significant differences at the extremes. *Proc Natl Acad Sci U S A*. 2015;112:10177–84.
- Cole LW, Guo W, Mower JP, Palmer JD. High and variable rates of repeat-mediated mitochondrial genome rearrangement in a genus of plants. *Mol Biol Evol*. 2018;35:2773–85.
- Sanchez-Puerta MV, Ceriotti LF, Gatica-Soria LM, Roulet ME, Garcia LE, Sato HA. Beyond parasitic convergence: unraveling the evolution of the organellar genomes in holoparasites. *Ann Bot*. 2023;32:909–28.
- Wang J, Kan S, Liao X, Zhou J, Tembrock LR, Daniell H, et al. Plant organellar genomes: much done, much more to do. *Trends Plant Sci*. 2024;29:754–69.
- Schellkunov MI, Nuraliev MS, Logacheva MD. Genomic comparison of non-photosynthetic plants from the family Balanophoraceae with their photosynthetic relatives. *PeerJ*. 2021;9:e12106.
- Ceriotti LF, Gatica-Soria L, Sanchez-Puerta MV. Cytonuclear coevolution in a holoparasitic plant with highly disparate organellar genomes. *Plant Mol Biol*. 2022;109:673–88.
- Yu R, Sun C, Zhong Y, Liu Y, Sanchez-Puerta MV, Mower JP, et al. The minicircular and extremely heteroplasmic mitogenome of the holoparasitic plant *Rhopalocnemis phalloides*. *Curr Biol*. 2022;32:470–9.
- Wang J, Zou Y, Mower JP, Reeve W, Wu Z. Rethinking the mutation hypotheses of plant organellar DNA. *Genomics Communications*. 2024;1:e003.
- Zwonitzer KD, Tressel LG, Wu Z, Kan S, Broz AK, Mower JP, et al. Genome copy number predicts extreme evolutionary rate variation in plant mitochondrial DNA. *Proc Natl Acad Sci U S A*. 2024;121:e2317240121.
- Molina J, Hazzouri KM, Nickrent D, Geisler M, Meyer RS, Pentony MM, et al. Possible loss of the chloroplast genome in the parasitic flowering plant *Rafflesia lagascae* (Rafflesiaceae). *Mol Biol Evol*. 2014;31:793–803.
- Cai L, Arnold BJ, Xi Z, Khost DE, Patel N, Hartmann CB, et al. Deeply altered genome architecture in the endoparasitic flowering plant *Sapria himalayana* Griff. (Rafflesiaceae). *Curr Biol*. 2021; 31: 1–10.
- Banerjee A, Stefanović S. A comparative study across the parasitic plants of *Cuscuta* subgenus *Grammica* (Convolvulaceae) reveals a possible loss of the plastid genome in its section *Subulatae*. *Planta*. 2023;257:66.
- Yu R, Jost M, Wanke S, Bruy D, Li P, Nickrent DL, et al. Evidence for plastome loss in the holoparasitic Mystropetalaceae. *bioRxiv*. 2025. <https://doi.org/10.1101/2025.07.03.662983>.
- Xi Z, Wang Y, Bradley RK, Sugumaran M, Marx CJ, Rest JS, et al. Massive mitochondrial gene transfer in a parasitic flowering plant clade. *PLoS Genet*. 2013;9:e1003265.
- Guo X, Hu X, Li J, Shao B, Wang Y, Wang L, et al. The *Sapria himalayana* genome provides new insights into the lifestyle of endoparasitic plants. *BMC Biol*. 2023;21(1):134.
- Roulet ME, Ceriotti LF, Gatica-Soria L, Sanchez-Puerta MV. Horizontally transferred mitochondrial DNA tracts become circular by microhomology-mediated repair pathways. *New Phytol*. 2024;243:2442–56.
- Kuijt J. The biology of parasitic flowering plants. Berkeley: University of California Press; 1969.
- Hansen B. The genus *Balanophora* J. R. & G.: Forster a taxonomic monograph. *Dansk Botanisk Arkiv*. 1972; 28: 1–188.
- Sanchez-Puerta MV, Edera A, Gandini CL, Williams AV, Howell KA, Nevill PG, et al. Genome-scale transfer of mitochondrial DNA from legume hosts to the holoparasite *Lophophytum mirabile* (Balanophoraceae). *Mol Phylogenet Evol*. 2019;132:243–50.
- Gatica-Soria LM, Roulet ME, Tulle WD, Sato HA, Barrandeguy ME, Sanchez-Puerta MV. Highly variable mitochondrial chromosome content in a holoparasitic plant due to recurrent gains of foreign circular DNA. *Physiol Plant*. 2025;177:e70231.
- Garcia LE, Edera AJ, Palmer JD, Sato H, Sanchez-Puerta MV. Horizontal gene transfers dominate the functional mitochondrial gene space of a holoparasitic plant. *New Phytol*. 2021;229:1701–14.
- Roulet ME, Garcia LE, Gandini CL, Sato H, Ponce G, Sanchez-Puerta MV. Multichromosomal structure and foreign tracts in the *Ombrophytum subterraneum* (Balanophoraceae) mitochondrial genome. *Plant Mol Biol*. 2020;103:623–38.
- Zhou S, Wei N, Jost M, Wanke S, Rees M, Liu Y, et al. The mitochondrial genome of the holoparasitic plant *Thonningia sanguinea* provides insights into the evolution of the multichromosomal structure. *Genome Biol Evol*. 2023;15:eivad155.
- Su H, Barkman TJ, Hao W, Jones SS, Naumann J, Skippington E, et al. Novel genetic code and record-setting AT-richness in the highly reduced plastid genome of the holoparasitic plant *Balanophora*. *Proc Natl Acad Sci U S A*. 2019;116:934–43.
- Chen X, Fang D, Wu C, Liu B, Liu Y, Sahu SK, et al. Comparative plastome analysis of root- and stem-feeding parasites of Santalales untangle the footprints of feeding mode and lifestyle transitions. *Genome Biol Evol*. 2020;12:3663–76.
- Ceriotti LF, Gatica-Soria LM, Guzman S, Sato HA, Tovar Luque E, Gonzalez MA, et al. The evolution of the plastid genomes in the holoparasitic Balanophoraceae. *Proc Biol Sci*. 2025;292:20242011.
- Chen X, Fang D, Xu Y, Duan K, Yoshida S, Yang S, et al. *Balanophora* genomes display massively convergent evolution with other extreme holoparasites and provide novel insights into parasite–host interactions. *Nat Plants*. 2023;9:1627–42.
- Sanchez-Puerta MV, Garcia LE, Wohlfeiler J, Ceriotti LF. Unparalleled replacement of native mitochondrial genes by foreign homologs in a holoparasitic plant. *New Phytol*. 2017;214:376–87.
- Warren JM, Sloan DB. Interchangeable parts: the evolutionarily dynamic tRNA population in plant mitochondria. *Mitochondrion*. 2020;52:144–56.
- Richardson AO, Rice DW, Young GJ, Alverson AJ, Palmer JD. The “fossilized” mitochondrial genome of *Liriodendron tulipifera*: ancestral gene content and order, ancestral editing sites, and extraordinarily low mutation rate. *BMC Biol*. 2013;11:29.

32. Sloan DB, Alverson AJ, Chuckalovcak JP, Wu M, McCauley DE, Palmer JD, et al. Rapid evolution of enormous, multichromosomal genomes in flowering plant mitochondria with exceptionally high mutation rates. *PLoS Biol.* 2012;10:e1001241.
33. Skippington E, Barkman TJ, Rice DW, Palmer JD. Miniaturized mitogenome of the parasitic plant *Viscum scurruloideum* is extremely divergent and dynamic and has lost all *nad* genes. *Proc Natl Acad Sci U S A.* 2015;112:E3515–24.
34. Edera AA, Howell KA, Nevill PG, Small I, Sanchez-Puerta MV. Evolution of *cox2* introns in angiosperm mitochondria and efficient splicing of an elongated *cox2i691* intron. *Gene.* 2023;869:147393.
35. Ngu M, Massel K, Bonen L. Group II introns in wheat mitochondria have degenerate structural features and varied splicing pathways. *Int J Biochem Cell Biol.* 2017;91:156–67.
36. Zanolungo S, Quiñones V, Moenne A, Holuigue L, Jordana X. Splicing and editing of *rps10* transcripts in potato mitochondria. *Curr Genet.* 1995;27:565–71.
37. Mower JP. Variation in protein gene and intron content among land plant mitogenomes. *Mitochondrion.* 2020;53:203–13.
38. Sanchez-Puerta MV, Cho Y, Mower JP, Alverson AJ, Palmer JD. Frequent, phylogenetically local horizontal transfer of the *cox1* group I intron in flowering plant mitochondria. *Mol Biol Evol.* 2008;25:1762–77.
39. Dong S, Zhao C, Chen F, Liu Y, Zhang S, Wu H, et al. The complete mitochondrial genome of the early flowering plant *Nymphaea colorata* is highly repetitive with low recombination. *BMC Genomics.* 2018;19:614.
40. Yu R, Sun C, Liu Y, Zhou R. Shifts from *cis*- to *trans*-splicing of five mitochondrial introns in *Tolypanthus maclurei*. *PeerJ.* 2021;9:e12260.
41. Wissinger B, Hiesel R, Schuster W, Brennicke A. The NADH-dehydrogenase subunit 5 gene in *Oenothera* mitochondria contains two introns and is co-transcribed with the 5S rRNA gene. *Mol Gen Genet.* 1988;212:56–65.
42. Lippok B, Brennicke A, Wissinger B. Differential RNA editing in closely related introns in *Oenothera* mitochondria. *Mol Gen Genet.* 1994;243:39–46.
43. Farré JC, Araya A. RNA splicing in higher plant mitochondria: determination of functional elements in group II intron from a chimeric *cox II* gene in electroporated wheat mitochondria. *Plant J.* 2002;29:203–13.
44. Sultan LD, Milesina D, Grewe F, Rolle K, Abudraham S, Glodowicz P, et al. The reverse transcriptase/RNA maturase protein matR is required for the splicing of various group II introns in Brassicaceae mitochondria. *Plant Cell.* 2016;28:2805–29.
45. Knoop V, Schuster W, Wissinger B, Brennicke A. Trans splicing integrates an exon of 22 nucleotides into the *nad5* mRNA in higher plant mitochondria. *EMBO J.* 1991;10:3483–93.
46. Pruitt KD, Hanson MR. Splicing of the *Petunia* cytochrome oxidase subunit II intron. *Curr Genet.* 1991;19:191–7.
47. Wissinger B, Schuster W, Brennicke A. Trans splicing in *Oenothera* mitochondria: *nad1* mRNAs are edited in exon and trans-splicing group II intron sequences. *Cell.* 1991;65:473–82.
48. Binder S, Marchfelder A, Brennicke A, Wissinger B. RNA editing in trans-splicing intron sequences of *nad2* mRNAs in *Oenothera* mitochondria. *J Biol Chem.* 1992;267:7615–23.
49. Knoop V, Altwasser M, Brennicke A. A tripartite group II intron in mitochondria of an angiosperm plant. *Mol Gen Genet.* 1997;255:269–76.
50. Malek O, Brennicke A, Knoop V. Evolution of trans-splicing plant mitochondrial introns in pre-Permian times. *Proc Natl Acad Sci U S A.* 1997;94:553–8.
51. Albertazzi FJ, Kudla J, Bock R. The *cox2* locus of the primitive angiosperm plant *Acorus calamus*: molecular structure, transcript processing and RNA editing. *Mol Gen Genet.* 1998;259:591–600.
52. Malek O, Knoop V. Trans-splicing group II introns in plant mitochondria: the complete set of cis-arranged homologs in ferns, fern allies, and a hornwort. *RNA.* 1998;4:1599–609.
53. Morawala-Patell V, Gualberto JM, Lamattina L, Grienemberger JM, Bonnard G. Cis- and trans-splicing and RNA editing are required for the expression of *nad2* in wheat mitochondria. *Mol Gen Genet.* 1998;258:503–11.
54. Ahlert D, Piepenburg K, Kudla J, Bock R. Evolutionary origin of a plant mitochondrial group II intron from a reverse transcriptase/maturase-encoding ancestor. *J Plant Res.* 2006;119:363–71.
55. Bégu D, Araya A. The horsetail *Equisetum arvense* mitochondria share two group I introns with the liverwort *Marchantia*, acquired a novel group II intron but lost intron-encoded ORFs. *Curr Genet.* 2009;55:69–79.
56. Regina TM, Quagliariello C. Lineage-specific group II intron gains and losses of the mitochondrial *rps3* gene in gymnosperms. *Plant Physiol Biochem.* 2010;48:646–54.
57. Volkmar U, Groth-Malonek M, Heinrichs J, Muhle H, Polsakiewicz M, Knoop V. Exclusive conservation of mitochondrial group II intron *nad4i548* among liverworts and its use for phylogenetic studies in this ancient plant clade. *Plant Biol (Stuttg).* 2012;14:382–91.
58. Knie N, Fischer S, Grewe F, Polsakiewicz M, Knoop V. Horsetails are the sister group to all other monilophytes and Marattiales are sister to leptosporangiate ferns. *Mol Phylogenet Evol.* 2015;90:140–9.
59. Zumkeller SM, Knoop V, Knie N. Convergent evolution of fern-specific mitochondrial group II intron *atp1i361g2* and its ancient source paralog *rps3i249g2* and independent losses of intron and RNA editing among Pteridaceae. *Genome Biol Evol.* 2016;8:2505–19.
60. Bonen L. Cis- and trans-splicing of group II introns in plant mitochondria. *Mitochondrion.* 2008;8:26–34.
61. Wright F. The effective number of codons used in a gene. *Gene.* 1990;87:23–9.
62. Xia H, Zhao W, Shi Y, Wang XR, Wang B. Microhomologies are associated with tandem duplications and structural variation in plant mitochondrial genomes. *Genome Biol Evol.* 2020;12:1965–74.
63. Sloan DB, Taylor DR. Testing for selection on synonymous sites in plant mitochondrial DNA: the role of codon bias and RNA editing. *J Mol Evol.* 2010;70:479–91.
64. Longley MJ, Nguyen D, Kunkel TA, Copeland WC. The fidelity of human DNA polymerase gamma with and without exonucleolytic proofreading and the p55 accessory subunit. *J Biol Chem.* 2001;276:38555–62.
65. McVey M, Lee SE. MMEJ repair of double-strand breaks (director's cut): deleted sequences and alternative endings. *Trends Genet.* 2008;24:529–38.
66. Yu R, Chen X, Long L, Jost M, Zhao R, Liu L, et al. De novo assembly and comparative analyses of mitochondrial genomes in Piperales. *Genome Biol Evol.* 2023;15:evad041.
67. Duchêne S, Ho SY, Holmes EC. Declining transition/transversion ratios through time reveal limitations to the accuracy of nucleotide substitution models. *BMC Evol Biol.* 2015;15:36.
68. Sloan DB, Wu Z. History of plastid DNA insertions reveals weak deletion and at mutation biases in angiosperm mitochondrial genomes. *Genome Biol Evol.* 2014;6:3210–21.
69. Cho Y, Mower JP, Qiu YL, Palmer JD. Mitochondrial substitution rates are extraordinarily elevated and variable in a genus of flowering plants. *Proc Natl Acad Sci U S A.* 2004;101:17741–6.
70. Parkinson CL, Mower JP, Qiu YL, Shirk AJ, Song K, Young ND, et al. Multiple major increases and decreases in mitochondrial substitution rates in the plant family Geraniaceae. *BMC Evol Biol.* 2005;5:73.
71. Zervas A, Petersen G, Seberg O. Mitochondrial genome evolution in parasitic plants. *BMC Evol Biol.* 2019;19:87.
72. Zaegel V, Guermann B, Le Ret M, Andrés C, Meyer D, Erhardt M, et al. The plant-specific ssDNA binding protein OSB1 is involved in the stoichiometric transmission of mitochondrial DNA in *Arabidopsis*. *Plant Cell.* 2006;18:35–63.
73. Shedge V, Arrieta-Montiel M, Christensen AC, Mackenzie SA. Plant mitochondrial recombination surveillance requires unusual *RecA* and *MutS* homologs. *Plant Cell.* 2007;19:1251–64.
74. Rowan BA, Oldenburg DJ, Bendich AJ. RecA maintains the integrity of chloroplast DNA molecules in *Arabidopsis*. *J Exp Bot.* 2010;61:2575–88.
75. Odahara M, Masuda Y, Sato M, Wakazaki M, Harada C, Toyooka K, et al. RECG maintains plastid and mitochondrial genome stability by suppressing extensive recombination between short dispersed repeats. *PLoS Genet.* 2015;11:e1005080.
76. Odahara M, Inouye T, Nishimura Y, Sekine Y. RECA plays a dual role in the maintenance of chloroplast genome stability in *Physcomitrella patens*. *Plant J.* 2015;84:516–26.
77. Odahara M, Sekine Y. RECX interacts with mitochondrial RECA to maintain mitochondrial genome stability. *Plant Physiol.* 2018;17:300–10.
78. Wallet C, Le Ret M, Bergdoll M, Bichara M, Dietrich A, Gualberto JM. The RECG1 DNA translocase is a key factor in recombination surveillance, repair, and segregation of the mitochondrial DNA in *Arabidopsis*. *Plant Cell.* 2015;27:2907–25.

79. Gualberto JM, Newton KJ. Plant mitochondrial genomes: dynamics and mechanisms of mutation. *Annu Rev Plant Biol.* 2017;68:225–52.
80. Chevigny N, Schatz-Daas D, Lotfi F, Gualberto JM. DNA repair and the stability of the plant mitochondrial genome. *Int J Mol Sci.* 2020;21:328.
81. Wu Z, Waneka G, Broz AK, King CR, Sloan DB. MSH1 is required for maintenance of the low mutation rates in plant mitochondrial and plastid genomes. *Proc Natl Acad Sci USA.* 2020;117:16448–55.
82. Waneka G, Broz AK, Wold-McGimsey F, Zou Y, Wu Z, Sloan DB. Disruption of recombination machinery alters the mutational landscape in plant organellar genomes. *G3: Genes, Genomes, Genetics.* 2025;15:jka029.
83. Havird JC, Trapp P, Miller CM, Bazos I, Sloan DB. Causes and consequences of rapidly evolving mtDNA in a plant lineage. *Genome Biol Evol.* 2017;9:323–36.
84. Banas AK, Hermanowicz P, Sztatelman O, Labuz J, Aggarwal C, Zglobicki P, et al. 6,4-PP photolyase encoded by *AtUVR3* is localized in nuclei, chloroplasts and mitochondria and its expression is down-regulated by light in a photosynthesis-dependent manner. *Plant Cell Physiol.* 2018;59:44–57.
85. Boesch P, Ibrahim N, Paulus F. Plant mitochondria possess a short-patch base excision DNA repair pathway. *Nucleic Acids Res.* 2009;37:5690–700.
86. Carrie C, Kühn K, Murcha MW. Approaches to defining dual-targeted proteins in *Arabidopsis*. *Plant J.* 2009;57:1128–39.
87. Chevigny N, Weber-Lotfi F, Le Blevenec A, Nadiaras C, Fertet A, Bichara M, et al. RADA-dependent branch migration has a predominant role in plant mitochondria and its defect leads to mtDNA instability and cell cycle arrest. *PLoS Genet.* 2022;18:e1010202.
88. Córdoba-Cañero D, Roldán-Arjona T, Ariza RR. *Arabidopsis* ZDP DNA 3'-phosphatase and ARP endonuclease function in 8-oxoG repair initiated by FPG and OGG 1 DNA glycosylases. *Plant J.* 2014;79:824–34.
89. Christensen AC, Lyznik A, Mohammed S, Elowsky CG, Elo A, Yule R, et al. Dual-domain, dual-targeting organellar protein presequences in *Arabidopsis* can use non-AUG start codons. *Plant Cell.* 2005;17:2805–16.
90. Dündar G, Teranishi M, Hidema J. Autophagy-deficient *Arabidopsis* mutant atg5, which shows ultraviolet-B sensitivity, cannot remove ultraviolet-B-induced fragmented mitochondria. *Photochem Photobiol Sci.* 2020;19:1717–29.
91. Edmondson AC, Song D, Alvarez LA, Wall MK, Almond D, McClellan DA, et al. Characterization of a mitochondrially targeted single-stranded DNA-binding protein in *Arabidopsis thaliana*. *Mol Genet Genomics.* 2005;273:115–22.
92. Ferrando B, Furlanetto ALDM, Gredilla R, Havelund JF, Hebelstrup KH, Möller IM, et al. DNA repair in plant mitochondria – a complete base excision repair pathway in potato tuber mitochondria. *Physiol Plant.* 2019;166:494–512.
93. Fuchs P, Rugen N, Carrie C, Elsässer M, Finkemeier I, Giese J, et al. Single organelle function and organization as estimated from *Arabidopsis* mitochondrial proteomics. *Plant J.* 2020;101:420–41.
94. García-Medel PL, Peralta-Castro A, Baruch-Torres N, Fuentes-Pascacio A, Pedroza-García JA, Cruz-Ramírez A, et al. *Arabidopsis thaliana* PrimPol is a primase and lesion bypass DNA polymerase with the biochemical characteristics to cope with DNA damage in the nucleus, mitochondria, and chloroplast. *Sci Rep.* 2021;11:20582.
95. Gutman BL, Niyogi KK. Evidence for base excision repair of oxidative DNA damage in chloroplasts of *Arabidopsis thaliana*. *J Biol Chem.* 2009;284:17006–12.
96. Hedtke B, Börner T, Weihe A. One RNA polymerase serving two genomes. *EMBO Rep.* 2000;1:435–40.
97. Janicka S, Kühn K, Le Ret M, Bonnard G, Imbault P, Augustyniak H, et al. A RAD52-like single-stranded DNA binding protein affects mitochondrial DNA repair by recombination. *Plant J.* 2012;72:423–35.
98. Khazi FR, Edmondson AC, Nielsen BL. An *Arabidopsis* homologue of bacterial RecA that complements an *E. coli* recA deletion is targeted to plant mitochondria. *Mol Genet Genomics.* 2003;269:454–63.
99. Klein T, Lockhart P, Batschauer A. An *Arabidopsis* protein closely related to *Synechocystis* cryptochrome is targeted to organelles. *Plant J.* 2003;35:93–103.
100. Kobayashi Y, Misumi O, Odahara M, Ishibashi K, Hirono M, Hidaka K, et al. Holliday junction resolvases mediate chloroplast nucleoid segregation. *Science.* 2017;356:631–4.
101. Krause K, Kilbiński I, Mulisch M, Rödiger A, Schäfer A, Krupinska K. DNA-binding proteins of the Whirly family in *Arabidopsis thaliana* are targeted to the organelles. *FEBS Lett.* 2005;579:3707–12.
102. Mahest HB, Subba P, Advani J, Shirke MD, Loganathan RM, Chandana SL, et al. Multi-omics driven assembly and annotation of the sandalwood (*Santalum album*) genome. *Plant Physiol.* 2018;176:2772–88.
103. Majeran W, Friso G, Asakura Y, Qu X, Huang M, Ponnala L, et al. Nucleoid-enriched proteomes in developing plastids and chloroplasts from maize leaves: a new conceptual framework for nucleoid functions. *Plant Physiol.* 2012;158:156–89.
104. Melonek J, Oetke S, Krupinska K. Multifunctionality of plastid nucleoids as revealed by proteome analyses. *Biochim Biophys Acta.* 2016;1864:1016–38.
105. Otake M, Teranishi M, Komatsu C, Hara M, Yoshiyama KO, Hidema J. Poaceae plants transfer cyclobutane pyrimidine dimer photolyase to chloroplasts for ultraviolet-B resistance. *Plant Physiol.* 2024;195:326–42.
106. Pfalz J, Liere K, Kandlbinder A, Dietz KJ, Oelmüller R. pTAC2, -6, and -12 are components of the transcriptionally active plastid chromosome that are required for plastid gene expression. *Plant Cell.* 2006;18:176–97.
107. Prasad M, Kataria P, Ningaraju S, Buddidathi R, Bankapalli K, Swetha C, et al. Double DJ-1 domain containing Arabidopsis DJ-1D is a robust macromolecule deglycase. *New Phytol.* 2022;236:1061–74.
108. Qian J, Zheng M, Wang L, Song Y, Yan J, Hsu YF. *Arabidopsis* mitochondrial single-stranded DNA-binding proteins SSB1 and SSB2 are essential regulators of mtDNA replication and homologous recombination. *J Integr Plant Biol.* 2022;64:1952–65.
109. Samach A, Melamed-Bessudo C, Avivi-Ragolski N, Pietrokovski S, Levy AA. Identification of plant RAD52 homologs and characterization of the *Arabidopsis thaliana* RAD52-like genes. *Plant Cell.* 2011;23:4266–79.
110. Schatz D, Le Blevenec A, Moratti FG, Chung KP, Mercier P, Iqbal RK, et al. R-loop control and mitochondrial genome stability require the 5'-3' exonuclease/flap endonuclease OEX1. *Plant Cell.* 2025;37:koaf104.
111. Sunderland PA, West CE, Waterworth WM, Bray CM. An evolutionarily conserved translation initiation mechanism regulates nuclear or mitochondrial targeting of DNA ligase 1 in *Arabidopsis thaliana*. *Plant J.* 2006;47:356–67.
112. Tarasenko VI, Katyshev AI, Yakovleva TV, Garnik EY, Chernikova VV, Konstantinov YM, et al. RPOtMp, an *Arabidopsis* RNA polymerase with dual targeting, plays an important role in mitochondria, but not in chloroplasts. *J Exp Bot.* 2016;67:5657–69.
113. Torres JR, Lopez IL, Ayala AM, Alvarez ME. The Arabidopsis DNA glycosylase MBD4L repairs the nuclear genome in vivo. *Plant J.* 2023;115:1633–46.
114. Wall MK, Mitchenall LA, Maxwell A. *Arabidopsis thaliana* DNA gyrase is targeted to chloroplasts and mitochondria. *Proc Natl Acad Sci U S A.* 2004;101:7821–6.
115. Xu CQ, Liu H, Zhou SS, Zhang DX, Zhao W, Wang S, et al. Genome sequence of *Malania oleifera*, a tree with great value for nervonic acid production. *Gigascience.* 2019;8:gij164.
116. Xu L, Carrie C, Law SR, Murcha MW, Whelan J. Acquisition, conservation, and loss of dual-targeted proteins in land plants. *Plant Physiol.* 2013;161:644–62.
117. Xu Y-Z, Arrieta-Montiel MP, Virdi KS, de Paula WB, Widhalm JR, Basset GJ, et al. MutS HOMOLOG1 is a nucleoid protein that alters mitochondrial and plastid properties and plant response to high light. *Plant Cell.* 2011;23:3428–41.
118. Yang Z, Hou Q, Cheng L, Xu W, Hong Y, Li S, et al. Rnase H1 cooperates with DNA gyrase to restrict R-loops and maintain genome integrity in *Arabidopsis* chloroplasts. *Plant Cell.* 2017;29:2478–97.
119. Zhang X, Li M, Bian Z, Chen X, Li Y, Xiong Y, et al. Improved chromosome-level genome assembly of Indian sandalwood (*Santalum album*). *Sci Data.* 2023;10:921.
120. Wafula EK, Zhang H, Von Kuster G, Leebens-Mack JH, Honaas LA, dePamphilis CW. PlantTribes2: Tools for comparative gene family analysis in plant genomics. *Front Plant Sci.* 2023;13:1011199.
121. Ceriotti LF, Roulet ME, Sanchez-Puerta MV. Plastomes in the holoparasitic family Balanophoraceae: extremely high AT content, severe gene content reduction, and two independent genetic code changes. *Mol Phylogenet Evol.* 2021;162:107208.

122. Schelkunov MI, Nuraliev MS, Logacheva MD. *Rhopalocnemis phalloides* has one of the most reduced and mutated plastid genomes known. *PeerJ*. 2019;7:e7500.
123. Kim W, Lautenschläger T, Bolin JF, Rees M, Nzuzi A, Zhou R, et al. Extreme plastomes in holoparasitic Balanophoraceae are not the norm. *BMC Genomics*. 2023;24:330.
124. Gandini C, García LE, Abbona CC, Sanchez-Puerta MV. The complete organelle genomes of *Physochlaina orientalis*: insights into short sequence repeats across seed plant mitochondrial genomes. *Mol Phylogenet Evol*. 2019;137:274–84.
125. Liu H, Zhao W, Zhang R-G, Mao J-F, Wang X-R. Repetitive elements, sequence turnover and cyto-nuclear gene transfer in gymnosperm mitogenomes. *Front Genet*. 2022;13:867736.
126. Smith DR, Hua J, Archibald JM, Lee RW. Palindromic genes in the linear mitochondrial genome of the nonphotosynthetic green alga *Polytoma magna*. *Genome Biol Evol*. 2013;5:1661–7.
127. Lee DK, Hsiang T, Lachance MA, Smith DR. The strange mitochondrial genomes of *Metschnikowia* yeasts. *Curr Biol*. 2020;30:R800–1.
128. van Beveren F, Eme L, López-García P, Ciobanu M, Moreira D. Independent size expansions and intron proliferation in red algal plastid and mitochondrial genomes. *Genome Biol Evol*. 2022; 14: evac037.
129. Liu W, Cai Y, Zhang Q, Shu F, Chen L, Ma X, et al. Subchromosome-scale nuclear and complete mitochondrial genome characteristics of *Morchella crassipes*. *Int J Mol Sci*. 2020;21:483.
130. Del Vasto M, Figueroa-Martínez F, Featherston J, González MA, Reyes-Prieto A, Durand PM, et al. Massive and widespread organelle genomic expansion in the green algal genus *Dunaliella*. *Genome Biol Evol*. 2015;7:656–63.
131. Guo W, Zhu A, Fan W, Mower JP. Complete mitochondrial genomes from the ferns *Ophioglossum californicum* and *Psilotum nudum* are highly repetitive with the largest organellar introns. *New Phytol*. 2017;213:391–403.
132. Smith DR. Updating our view of organelle genome nucleotide landscape. *Front Genet*. 2012;3:175.
133. Nguyen DT, Wu B, Xiao S, Hao W. Evolution of a record-setting AT-rich genome: indel mutation, recombination, and substitution bias. *Genome Biol Evol*. 2020;12:2344–54.
134. Tang J-Y, Wei R, Zhang X-C, Xiang Q-P. Mitogenome-based phylogenomics provides insights into the positions of the enigmatic *sinensis* group and the *sanguinolenta* group in Selaginellaceae (Lycopphyte). *Mol Phylogenet Evol*. 2023;179:107673.
135. Figueroa-Martínez F, Nedelcu AM, Smith DR, Reyes-Prieto A. The plastid genome of *Polytoma uvella* is the largest known among colorless algae and plants and reflects contrasting evolutionary paths to nonphotosynthetic lifestyles. *Plant Physiol*. 2017;173:932–43.
136. Sloan DB, Oxelman B, Rautenberg A, Taylor DR. Phylogenetic analysis of mitochondrial substitution rate variation in the angiosperm tribe Sileneae. *BMC Evol Biol*. 2009;9:260.
137. Choi K, Weng ML, Ruhlman TA, Jansen RK. Extensive variation in nucleotide substitution rate and gene/intron loss in mitochondrial genomes of *Pelargonium*. *Mol Phylogenet Evol*. 2021;155:106986.
138. Mower JP, Touzet P, Gummow JS, Delph LF, Palmer JD. Extensive variation in synonymous substitution rates in mitochondrial genes of seed plants. *BMC Evol Biol*. 2007;7:135.
139. Álvarez-Escribano I, Sasse C, Bok JW, Na H, Amirebrahimi M, Lipzen A, et al. Genome sequencing of evolved *Aspergilli* populations reveals robust genomes, transversions in *A. flavus*, and sexual aberrancy in non-homologous end-joining mutants. *BMC Biol*. 2019;17:88.
140. Conrad DF, Keebler JE, DePristo MA, Lindsay SJ, Zhang Y, Casals F, et al. Variation in genome-wide mutation rates within and between human families. *Nat Genet*. 2011;43:712–4.
141. Denver DR, Dolan PC, Wilhelm LJ, Sung W, Lucas-Lledó JI, Howe DK, et al. A genome-wide view of *Caenorhabditis elegans* base-substitution mutation processes. *Proc Natl Acad Sci U S A*. 2009;106:16310–4.
142. Denver DR, Wilhelm LJ, Howe DK, Gafner K, Dolan PC, Baer CF. Variation in base-substitution mutation in experimental and natural lineages of *Caenorhabditis* nematodes. *Genome Biol Evol*. 2012;4:513–22.
143. Dillon MM, Sung W, Lynch M, Cooper VS. The rate and molecular spectrum of spontaneous mutations in the GC-rich multichromosome genome of *Burkholderia cenocepacia*. *Genetics*. 2015;200:935–46.
144. Dillon MM, Sung W, Sebra R, Lynch M, Cooper VS. Genome-wide biases in the rate and molecular spectrum of spontaneous mutations in *Vibrio cholerae* and *Vibrio fischeri*. *Mol Biol Evol*. 2017;34:93–109.
145. Farlow A, Long H, Arnoux S, Sung W, Doak TG, Nordborg M, et al. The spontaneous mutation rate in the fission yeast *Schizosaccharomyces pombe*. *Genetics*. 2015;201:737–44.
146. Hamilton WL, Claessens A, Otto TD, Kekre M, Fairhurst RM, Rayner JC, et al. Extreme mutation bias and high AT content in *Plasmodium falciparum*. *Nucleic Acids Res*. 2017;45:1889–901.
147. Keith N, Tucker AE, Jackson CE, Sung W, Lucas Lledó JI, Schrider DR, et al. High mutational rates of large-scale duplication and deletion in *Daphnia pulex*. *Genome Res*. 2016;26:60–9.
148. Kong A, Frigge ML, Masson G, Besenbacher S, Sulem P, Magnusson G, et al. Rate of de novo mutations and the importance of father's age to disease risk. *Nature*. 2012;488:471–5.
149. Kucukyildirim S, Long H, Sung W, Miller SF, Doak TG, Lynch M. The rate and spectrum of spontaneous mutations in *Mycobacterium smegmatis*, a bacterium naturally devoid of the postreplicative mismatch repair pathway. *G3 Genes[Genomes]Genetics*. 2016;6:2157–63.
150. Long H, Behringer MG, Williams E, Te R, Lynch M. Similar mutation rates but highly diverse mutation spectra in Ascomycete and Basidiomycete yeasts. *Genome Biol Evol*. 2016;8:3815–21.
151. Long H, Kucukyildirim S, Sung W, Williams E, Lee H, Ackerman M, et al. Background mutational features of the radiation-resistant bacterium *Deinococcus radiodurans*. *Mol Biol Evol*. 2015;32:2383–92.
152. Long H, Miller SF, Strauss C, Zhao C, Cheng L, Ye Z, et al. Antibiotic treatment enhances the genome-wide mutation rate of target cells. *Proc Natl Acad Sci U S A*. 2016;113:E2498–505.
153. Long H, Sung W, Kucukyildirim S, Williams E, Miller SF, Guo W, et al. Evolutionary determinants of genome-wide nucleotide composition. *Nat Ecol Evol*. 2018;2(2):237–40.
154. Lynch M, Ackerman MS, Gout JF, Long H, Sung W, Thomas WK, et al. Genetic drift, selection and the evolution of the mutation rate. *Nat Rev Genet*. 2016;17:704–14.
155. Ness RW, Morgan AD, Vasanthakrishnan RB, Colegrave N, Keightley PD. Extensive de novo mutation rate variation between individuals and across the genome of *Chlamydomonas reinhardtii*. *Genome Res*. 2015;25:1739–49.
156. Ossowski S, Schneeberger K, Lucas-Lledó JI, Warthmann N, Clark RM, Shaw RG, et al. The rate and molecular spectrum of spontaneous mutations in *Arabidopsis thaliana*. *Science*. 2010;327:92–4.
157. Schrider DR, Houle D, Lynch M, Hahn MW. Rates and genomic consequences of spontaneous mutational events in *Drosophila melanogaster*. *Genetics*. 2013;194:937–54.
158. Smeds L, Qvarnström A, Ellegren H. Direct estimate of the rate of germline mutation in a bird. *Genome Res*. 2016;26:1211–8.
159. Sun Y, Sun Y, Powell KE, Sung W, Lynch M, Moran MA, et al. Spontaneous mutations of a model heterotrophic marine bacterium. *ISME J*. 2017;11:1713–8.
160. Sung W, Ackerman MS, Dillon MM, Platt TG, Fuqua C, Cooper VS, et al. Evolution of the insertion-deletion mutation rate across the tree of life. *G3 Genes[Genomes]Genetics*. 2016;6:2583–91.
161. Sung W, Ackerman MS, Gout JF, Miller SF, Williams E, Foster PL, et al. Asymmetric context-dependent mutation patterns revealed through mutation-accumulation experiments. *Mol Biol Evol*. 2015;32:1672–83.
162. Sung W, Ackerman MS, Miller SF, Doak TG, Lynch M. Drift-barrier hypothesis and mutation-rate evolution. *Proc Natl Acad Sci USA*. 2012;109:18488–92.
163. Uchimura A, Higuchi M, Minakuchi Y, Ohno M, Toyoda A, Fujiyama A, et al. Germline mutation rates and the long-term phenotypic effects of mutation accumulation in wild-type laboratory mice and mutator mice. *Genome Res*. 2015;25:1125–34.
164. Venn O, Turner I, Mathieson I, de Groot N, Bontrop R, McVean G. Strong male bias drives germline mutation in chimpanzees. *Science*. 2014;344:1272–5.
165. Yang S, Wang L, Huang J, Zhang X, Yuan Y, Chen JQ, et al. Parent-progeny sequencing indicates higher mutation rates in heterozygotes. *Nature*. 2015;523(7523):463–7.
166. Zhu YO, Siegal ML, Hall DW, Petrov DA. Precise estimates of mutation rate and spectrum in yeast. *Proc Natl Acad Sci U S A*. 2014;111:E2310–8.

167. Payne JL, Menardo F, Trauner A, Borrell S, Gygli SM, Loiseau C, et al. Transition bias influences the evolution of antibiotic resistance in *Mycobacterium tuberculosis*. *PLoS Biol.* 2019;17:e3000265.
168. Iliushchenko D, Efmenko B, Mikhailova AG, Shamanskiy V, Saparbaev MK, Matkarimov BT, et al. Deciphering the foundations of mitochondrial mutational spectra: replication-driven and damage-induced signatures across chordate classes. *Mol Biol Evol.* 2025. <https://doi.org/10.1093/molbev/msae261>.
169. Huang J-L, Sun G-L, Zhang D-M. Molecular evolution and phylogeny of the angiosperm *ycf2* gene. *J Syst Evol.* 2010;48:240–8.
170. Zhang Y, Yang G, Fang M, Deng C, Zhang KQ, Yu Z, et al. Comparative analyses of mitochondrial genomes provide evolutionary insights into nematode-trapping fungi. *Front Microbiol.* 2020;11:617.
171. Carlson J, DeWitt WS, Harris K. Inferring evolutionary dynamics of mutation rates through the lens of mutation spectrum variation. *Curr Opin Genet Dev.* 2020;62:50–7.
172. Wu CS, Chaw SM. Large-scale comparative analysis reveals the mechanisms driving plastomic compaction, reduction, and inversions in Conifers II (Cupressophytes). *Genome Biol Evol.* 2016;8:3740–50.
173. Guo Y-Y, Yang J-X, Li H-K, Zhao H-S. Chloroplast genomes of two species of *Cypripedium*: expanded genome size and proliferation of AT-biased repeat sequences. *Front Plant Sci.* 2021;12:609729.
174. Rice DW, Alverson AJ, Richardson AO, Young GJ, Sanchez-Puerta MV, Munzinger J, et al. Horizontal transfer of entire genomes via mitochondrial fusion in the angiosperm *Amborella*. *Science.* 2013;342:1468–73.
175. Cai L. Rethinking convergence in plant parasitism through the lens of molecular and population genetic processes. *Am J Bot.* 2023;110:e16174.
176. DeTar RA, Chustecki JM, Martinez-Hottovy A, Ceriotti LF, Broz AK, Lou X, et al. Photosynthetic demands on translational machinery drive retention of redundant tRNA metabolism in plant organelles. *Proc Natl Acad Sci U S A.* 2024;121:e2421485121.
177. Schelkunov MI, Penin AA, Logacheva MD. RNA-seq highlights parallel and contrasting patterns in the evolution of the nuclear genome of fully mycoheterotrophic plants. *BMC Genomics.* 2018;19:602.
178. Wu CS, Chaw SM. Evolutionary stasis in cycad plastomes and the first case of plastome GC-biased gene conversion. *Genome Biol Evol.* 2015;7:2000–9.
179. Li FW, Kuo LY, Pryer KM, Rothfels CJ. Genes translocated into the plastid inverted repeat show decelerated substitution rates and elevated GC content. *Genome Biol Evol.* 2016;8:2452–8.
180. Niu Z, Xue Q, Wang H, Xie X, Zhu S, Liu W, et al. Mutational biases and GC-biased gene conversion affect GC content in the plastomes of *Dendrobium* genus. *Int J Mol Sci.* 2017;18:2307.
181. Ruhlman TA, Zhang J, Blazier JC, Sabir JSM, Jansen RK. Recombination-dependent replication and gene conversion homogenize repeat sequences and diversify plastid genome structure. *Am J Bot.* 2017;104:559–72.
182. Moran NA, McCutcheon JP, Nakabachi A. Genomics and evolution of heritable bacterial symbionts. *Annu Rev Genet.* 2008;42:165–90.
183. McCutcheon JP, Moran NA. Extreme genome reduction in symbiotic bacteria. *Nat Rev Microbiol.* 2012;10:13–26.
184. Kang JS, Zhang HR, Wang YR, Liang SQ, Mao ZY, Zhang XC, et al. Distinct evolutionary pattern of organelle genomes linked to the nuclear genome in Selaginellaceae. *Plant J.* 2020;104:1657–72.
185. Xiang PQ, Tang YJ, Yu GJ, Smith RD, Zhu MY, Wang RY, et al. The evolution of extremely diverged plastomes in Selaginellaceae (lycophyte) is driven by repeat patterns and the underlying DNA maintenance machinery. *Plant J.* 2022;111:768–84.
186. Lucas-Lledó JI, Lynch M. Evolution of mutation rates: phylogenomic analysis of the photolyase/cryptochrome family. *Mol Biol Evol.* 2009;26:1143–53.
187. Heide-Jørgensen HS. Parasitic flowering plants. Leiden: Brill NV; 2008.
188. Solovchenko AE, Merzlyak MN. Screening of visible and UV radiation as a photoprotective mechanism in plants. *Russ J Plant Physiol.* 2008;55:719–37.
189. Sharma M, Bennewitz B, Klösgen RB. Rather rule than exception? How to evaluate the relevance of dual protein targeting to mitochondria and chloroplasts. *Photosynth Res.* 2018;138:335–43.
190. Dunn CD, Paavilainen VO. Wherever i may roam: organellar protein targeting and evolvability. *Curr Opin Genet Dev.* 2019;58–59:9–16.
191. García-Medel PL, Baruch-Torres N, Peralta-Castro A, Trasviña-Arenas CH, Torres-Larios A, Briebe LG. Plant organellar DNA polymerases repair double-stranded breaks by microhomology-mediated end-joining. *Nucleic Acids Res.* 2019;47:3028–44.
192. Forsythe ES, Sharbrough J, Havird JC, Warren JM, Sloan DB. CyMIRA: The cytonuclear molecular interactions reference for *Arabidopsis*. *Genome Biol Evol.* 2019;11:2194–202.
193. Sloan DB, Alverson AJ, Wu M, Palmer JD, Taylor DR. Recent acceleration of plastid sequence and structural evolution coincides with extreme mitochondrial divergence in the angiosperm genus *Silene*. *Genome Biol Evol.* 2012;4:294–306.
194. Zou Y, Zhu W, Hou Y, Sloan DB, Wu Z. The evolutionary dynamics of organellar pan-genomes in *Arabidopsis thaliana*. *Genome Biol.* 2025;26:240.
195. Lee C, Ruhlman TA, Jansen RK. Rate accelerations in plastid and mitochondrial genomes of Cyperaceae occur in the same clades. *Mol Phylogenet Evol.* 2023;182:107760.
196. Nakabeppu Y. Cellular Levels of 8-Oxoguanine in either DNA or the Nucleotide Pool Play Pivotal Roles in Carcinogenesis and Survival of Cancer Cells. *Int J Mol Sci.* 2014;15:12543–57.
197. Liu F, Fan W, Yang JB, Xiang CL, Mower JP, Li DZ, et al. Episodic and guanine-cytosine-biased bursts of intragenomic and interspecific synonymous divergence in Ajugoideae (Lamiaceae) mitogenomes. *New Phytol.* 2020;228:1107–14.
198. Cameron SL, Yoshizawa K, Mizukoshi A, Whiting MF, Johnson KP. Mitochondrial genome deletions and minicircles are common in lice (Insecta: Phthiraptera). *BMC Genomics.* 2011;12:394.
199. Lee Y, Cho CH, Noh C, Yang JH, Park SI, Lee YM, et al. Origin of minicircular mitochondrial genomes in red algae. *Nat Commun.* 2023;14:3363.
200. Koren S, Walenz BP, Berlin K, Miller JR, Bergman NH, Phillippy AM. Canu: scalable and accurate long-read assembly via adaptive k-mer weighting and repeat separation. *Genome Res.* 2017;27:722–36.
201. Salmela L, Rivals E. Lordec: accurate and efficient long read error correction. *Bioinformatics.* 2014;30:3506–14.
202. Chaisson MJ, Tesler G. Mapping single molecule sequencing reads using Basic Local Alignment with Successive Refinement (BLASR): application and theory. *BMC Bioinform.* 2012;13:238.
203. Li H, Durbin R. Fast and accurate short read alignment with Burrows-Wheeler transform. *Bioinformatics.* 2009;25:1754–60.
204. Jin J, Yu WB, Yang JB, Song Y, dePamphilis CW, Yi TS, et al. Getorganelle: a fast and versatile toolkit for accurate de novo assembly of organelle genomes. *Genome Biol.* 2020;21:241.
205. Walker BJ, Abeel T, Shea T, Priest M, Abouelliel A, Sakthikumar S, et al. Pilon: an integrated tool for comprehensive microbial variant detection and genome assembly improvement. *PLoS ONE.* 2014;9:e112963.
206. Li H. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics.* 2018;34:3094–100.
207. Nurk S, Bankevich A, Antipov D, Gurevich AA, Korobeynikov A, Lapidus A, et al. Assembling genomes and mini-metagenomes from highly chimeric reads. *Res Comput Mol Biol.* 2013;20:714–37.
208. Dierckx N, Mardulyn P, Smits G. Novoplasty: de novo assembly of organelle genomes from whole genome data. *Nucleic Acids Res.* 2016;45:e18.
209. Hu J, Fan J, Sun Z, Liu S. Nextpolish: a fast and efficient genome polishing tool for long-read assembly. *Bioinformatics.* 2020;36:2253–5.
210. Skippington E, Barkman TJ, Rice DW, Palmer JD. Comparative mitogenomics indicates respiratory competence in parasitic *Viscum* despite loss of complex I and extreme sequence divergence, and reveals horizontal gene transfer and remarkable variation in genome size. *BMC Plant Biol.* 2017;17:49.
211. Lowe TM, Chan PP. tRNAscan-SE on-line: search and contextual analysis of transfer RNA genes. *Nucleic Acids Res.* 2016;44:W54–7.
212. Krzywinski M, Schein J, Birol I, Connors J, Gascoyne R, Horsman D, et al. Circos: an information aesthetic for comparative genomics. *Genome Res.* 2009;19:1639–45.
213. Li H, Handsaker B, Wysoker A, Fennell T, Ruan J, Homer N, et al. The sequence alignment/map (SAM) format and SAMtools. *Bioinformatics.* 2009;25:2078–9.
214. Katoh K, Rozewicki J, Yamada KD. Mafft online service: multiple sequence alignment, interactive sequence choice and visualization. *Brief Bioinform.* 2019;20:1160–6.

215. Wynn EL, Christensen AC. Repeats of unusual size in plant mitochondrial genomes: identification, incidence and evolution. *G3 Genes[Genomes]Genetics*. 2019;9:549–59.
216. Benson G. Tandem repeats finder: a program to analyze DNA sequences. *Nucleic Acids Res*. 1999;27:573–80.
217. Tempel S. Using and understanding RepeatMasker. In: Bigot Y, editor. *Mobile Genetic Elements. Methods in Molecular Biology (Methods and Protocols)*, vol 859. New Jersey: Humana Press; 2012. p. 29–51.
218. Edgar RC. MUSCLE: multiple sequence alignment with high accuracy and high throughput. *Nucleic Acids Res*. 2004;32:1792–7.
219. Castresana J. Selection of conserved blocks from multiple alignments for their use in phylogenetic analysis. *Mol Biol Evol*. 2000;17:540–52.
220. Edera AA, Gandini CL, Sanchez-Puerta MV. Towards a comprehensive picture of C-to-U RNA editing sites in angiosperm mitochondria. *Plant Mol Biol*. 2018;97:215–31.
221. Bell D, Lin Q, Gerelle WK, Joya S, Chang Y, Taylor ZN, et al. Organellar data sets confirm a cryptic consensus on (unrooted) land-plant relationships and provide new insights into bryophyte molecular evolution. *Am J Bot*. 2020;107(1):91–115.
222. Stamatakis A. RAxML version 8: a tool for phylogenetic analysis and post-analysis of large phylogenies. *Bioinformatics*. 2014;30:1312–3.
223. Peden JF. Analysis of codon usage. PhD Thesis. University of Nottingham; 2000.
224. Yang Z. PAML 4: a program package for phylogenetic analysis by maximum likelihood. *Mol Biol Evol*. 2007;24:1586–91.
225. Iv APG. An update of the Angiosperm Phylogeny Group classification for the orders and families of flowering plants: APG IV. *Bot J Linn Soc*. 2016;181:1–20.
226. Leebens-Mack JH, Barker MS, Carpenter EJ, Deyholos MK, Gitzendanner MA, Graham SW, et al. One thousand plant transcriptomes and the phylogenomics of green plants. *Nature*. 2019;574:679–85.
227. Wong GK, Soltis DE, Leebens-Mack J, Wickett NJ, Barker MS, Van de Peer Y, et al. Sequencing and analyzing the transcriptomes of a thousand species across the tree of life for green plants. *Annu Rev Plant Biol*. 2020;71:741–65.
228. Adams KL, Qiu YL, Stoutemyer M, Palmer JD. Punctuated evolution of mitochondrial gene content: high and variable rates of mitochondrial gene loss and transfer to the nucleus during angiosperm evolution. *Proc Natl Acad Sci U S A*. 2002;99:9905–12.
229. Pond SL, Frost SD, Muse SV. HyPhy: hypothesis testing using phylogenies. *Bioinformatics*. 2005;21:676–9.
230. Lynch M. The origins of genome architecture. Sunderland, MA: Sinauer Associates; 2007.
231. Grabherr MG, Haas BJ, Yassour M, Levin JZ, Thompson DA, Amit I, et al. Full-length transcriptome assembly from RNA-seq data without a reference genome. *Nat Biotechnol*. 2011;29:644–52.
232. Haas BJ, Papanicolaou A, Yassour M, Grabherr M, Blood PD, Bowden J, et al. De novo transcript sequence reconstruction from RNA-seq using the Trinity platform for reference generation and analysis. *Nat Protoc*. 2013;8:1494–512.
233. Fu L, Niu B, Zhu Z, Wu S, Li W. CD-HIT: accelerated for clustering the next-generation sequencing data. *Bioinformatics*. 2012;28(23):3150–2.
234. Simao FA, Waterhouse RM, Ioannidis P, Kriventseva EV, Zdobnov EM. BUSCO: assessing genome assembly and annotation completeness with single-copy orthologs. *Bioinformatics*. 2015;31:3210–2.
235. Sloan DB, Triant DA, Wu M, Taylor DR. Cytonuclear interactions and relaxed selection accelerate sequence evolution in organelle ribosomes. *Mol Biol Evol*. 2014;31:673–82.
236. Williams AM, Itgen MW, Broz AK, Carter OG, Sloan DB. Long-read transcriptome and other genomic resources for the angiosperm *Silene noctiflora*. *G3 (Bethesda)*. 2021; 11: jkab189.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH (“Springer Nature”).

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users (“Users”), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use (“Terms”). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval, sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com